Adventist University of Central Africa

TAX PROFESSIONAL MANAGEMENT SYSTEM

Case study: Rwanda Revenue Authority

A Final Year Project Presented in partial fulfillment of the requirements for the degree of BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

Major in

Software Engineering

By

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March, 2026

ABSTRACT

A final year project for a Bachelor's Degree in Science in Information Technology
Emphasis in software engineering
Adventist University of Central Africa

Title: Tax Professional Management system
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Date Completed: March 2026

The Tax Professional Management System (TPMS) is a web-based information system designed to support the Rwanda Revenue Authority (RRA) in the effective management of tax professionals and tax assistants. The system focuses on automating the processes of application, licensing, renewal, approval, rejection, and monitoring of tax professionals in accordance with national tax regulations and professional standards.

Currently, many tax professional management activities rely on manual procedures and fragmented digital tools, which lead to delays, data inconsistency, lack of transparency, and increased administrative workload. TPMS aims to address these challenges by providing a centralized digital platform where applicants can submit applications online, upload all required supporting documents, track application status in real time, and renew licenses upon expiry.

The system also provides an administrative portal for RRA officers to review submitted applications, verify compliance with legal and professional requirements, approve or reject applications, and generate detailed reports for monitoring and decision-making purposes. By implementing role-based access control, document validation, and automated workflows, TPMS enhances efficiency, accountability, data accuracy, and service delivery. The implementation of this system contributes to Rwanda's e-government strategy and strengthens regulatory oversight of tax professionals.

DECLARATION

ISHIMWE Aurore Nadine, Student ID Number **26420**, is a student at Adventist University of Central Africa in the Faculty of Information Technology, Department of software engineering. I do hereby declare that this research project report is entirely the real reflection of my own original work and experience to the best of my knowledge. It has never been either partially or wholly presented in any university or any higher learning institution.

Signature: ……...

Date: ……/…..…/……..

APPROVAL

I, **BISENGIMANA KAGIRANEZA Ivan,** hereby certify that this final year project has been done under my supervision and submitted with my approval.

Signature: ……

Date: ……/……/……

DEDICATION

I dedicate this final year project
To my parents, for their constant love,
prayers and support.

To my supervisor and all my lecturers for their guidance,
and pouring into me as much as they could.
To my beloved siblings,
classmates and friends, for their support, love,
prayers and for challenging me,
I appreciate you, thanks.

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LIST OF ABBREVIATIONS

API Application Programming Interface

AUCA Adventist University of Central Africa

BPMN Business Process Model and Notation

TPMS Tax Professional Management System

DBMS Database Management System

HTTP Hypertext Transfer Protocol

ICT Information and Communication Technology

IT Information Technology

ML Machine Learning

NGO Non-Governmental Organization

NPO Non-Profit Organization

RDBMS Relational Database Management System

SQL Structured Query Language

UI User Interface

UX User Experience

NST National Strategy for Transformation

alt alternative

opt optional

ACKNOWLEDGMENTS

Firstly, I would like to express my sincere gratitude to Almighty God for His blessings, guidance, and protection throughout this project. His grace has been my strength in every step of this journey, and I am truly thankful for His constant presence in my life.

My sincere thanks go to the academic staff of the Faculty of Information Technology, Department of software engineering, and the entire Administration at AUCA. Your support and the quality education I received have played a vital role in shaping my academic and professional development.

I extend my heartfelt appreciation to Mr. Bisengimana Kagiraneza Ivan , my supervisor, for his consistent guidance, patience, and support throughout this project. His expertise and constructive feedback were invaluable to the successful completion of this work.

I am deeply grateful to my parents for their unwavering love, support, prayers and encouragement. Your belief in me and your sacrifices have been the foundation of my success. Thank you for always being there for me.

To my siblings, thank you for your inspiration and continuous encouragement that pushed me to stay focused and achieve my goals.

To My Friends and Classmates at AUCA, I am thankful for their encouragement, collaboration, and companionship during my studies. Your support and friendship have made this academic journey enriching and memorable.

I want to express my sincere gratitude to the Almighty God for His blessings and guidance throughout this project. His grace and strength have carried me through every challenge, and for that, I am truly thankful.

May God bless you all abundantly.

ISHIMWE Aurore Nadine

CHAPTER 1

GENERAL INTRODUCTION

Introduction

In today's rapidly evolving digital environment, information systems play a crucial role in enhancing efficiency, transparency, and accountability within public and regulatory institutions. In tax administration, tax professionals serve as key intermediaries between tax adviser and tax authorities, contributing to compliance, accurate reporting, and effective revenue collection. The adoption of digital solutions is therefore essential for improving regulatory processes and service delivery.

In Rwanda, the management of tax adviser and tax authority largely relies on manual and semi-automated procedures that are often fragmented, inefficient, and lacking transparency. These limitations result in processing delays, difficulties in document verification, poor license tracking, and inconsistent record management. Such challenges increase administrative workload and negatively affect applicants through unclear procedures and limited visibility into application status.

The Tax Professional Management System (TPMS) is a web-based information system designed to streamline the registration, licensing, renewal, and monitoring of tax professionals at the Rwanda Revenue Authority (RRA). By integrating centralized data management, secure document handling, automated workflows, and role-based access control, the system enhances efficiency, accuracy, and transparency. Key functionalities include online applications, document upload and validation, approval workflows, license management, renewal tracking, and reporting tools.

The ultimate vision of TPMS is to establish a centralized and user-friendly digital platform that improves accessibility for applicants while enabling RRA officers to manage regulatory processes effectively. The system aligns with Rwanda's Vision 2050 and the National Strategy for Transformation (NST2, 2025–2031), which emphasize digital transformation and improved public service delivery.

Background of the Study

General Historical Background

The regulation and management of professional services have long been essential components of effective governance and institutional accountability worldwide. Across different sectors, regulatory bodies are responsible for ensuring that professionals meet established standards, comply with legal requirements, and maintain ethical practices. Traditionally, professional registration, licensing, and monitoring processes relied on manual procedures, including physical document submission, paper-based records, and face-to-face administrative interactions. While such approaches were manageable at smaller scales, they became increasingly inefficient as the number of professionals grew and regulatory requirements became more complex.

In recent years, digital transformation has significantly reshaped how professional management and regulatory processes are conducted globally. Advances in Information and Communication Technologies (ICT), centralized databases, and automated workflow systems have enabled institutions to digitize registration, licensing, compliance monitoring, and reporting activities. These digital systems improve efficiency, reduce administrative delays, enhance transparency, and strengthen data accuracy by replacing fragmented manual processes with integrated platforms (World Bank, 2022). As governments continue to adopt e-government strategies, technology-driven regulatory systems have become critical tools for modern public administration.

In Rwanda, professional regulation plays a crucial role in ensuring compliance, maintaining service quality, and strengthening institutional effectiveness. The Rwanda Revenue Authority (RRA), as the national body responsible for tax administration, regulates tax professionals and tax assistants to ensure that only qualified individuals provide tax-related advisory and compliance services. Tax professionals serve as key intermediaries between tax adviser and the tax authority, contributing to accurate tax reporting, regulatory compliance, and national revenue collection.

Historically, the management of tax professional applications, licensing, and renewals has relied on manual and semi-automated processes. Applicants often submit physical documents, while administrative activities such as verification, approval, record-keeping, and renewal

tracking are handled through paper files, spreadsheets, and email communication. These practices introduce challenges including processing delays, difficulties in document verification, inconsistent data management, and limited transparency in decision-making processes (MINICT, 2022).

Furthermore, the absence of a centralized digital platform limits the ability of regulatory officers to efficiently track license validity, monitor professional records, analyze trends, and generate reliable reports for planning and policy formulation. As the number of tax professionals continues to increase alongside the growing complexity of tax regulations, existing manual procedures struggle to ensure efficiency, accuracy, and effective regulatory oversight.

It is within this context that the Tax Professional Management System (TPMS) was conceptualized. This study contributes by proposing and developing a centralized web-based system designed to modernize the registration, licensing, renewal, and monitoring of tax professionals at the Rwanda Revenue Authority. TPMS replaces fragmented manual processes with automated application workflows, secure document management, real-time status tracking, license lifecycle management, and reporting functionalities.

Additionally, TPMS enhances transparency and accountability by providing structured approval workflows, centralized record management, audit trails, and analytical reporting tools that support evidence-based decision-making. The system enables both applicants and regulatory officers to interact through a secure, efficient, and user-friendly digital environment.

From an academic and practical perspective, this study demonstrates how digital technologies can be applied to address real-world challenges in regulatory and public sector administration. As a Bachelor's Degree project in Information Technology at the Adventist University of Central Africa (AUCA), TPMS represents both a technical innovation and an institutional efficiency solution. The system contributes to Rwanda's digital transformation agenda by improving service delivery, enhancing regulatory effectiveness, and strengthening data integrity within tax professional management.

Statement of the Problem

The main problem addressed in this study is the lack of an efficient, centralized, and automated system for managing tax professional registration, licensing, renewal, and monitoring processes at the Rwanda Revenue Authority (RRA). The proposed system, named the Tax Professional Management System (TPMS), seeks to address these challenges by modernizing and digitizing regulatory workflows. The system is designed to streamline administrative processes, centralize professional records, improve transparency, and enhance regulatory efficiency.

Manual Application and Registration Processes:
The existing reliance on manual and semi-manual procedures requires applicants to submit physical documents and forms. These processes are prone to errors such as incomplete submissions, duplicate records, misplaced documents, and inconsistencies in data capture.

Inefficient Document Verification and Management:
Physical document handling complicates verification procedures, increases processing time, and creates risks of document loss or damage. The absence of automated validation mechanisms makes compliance checks time-consuming and inconsistent.

Limited Application Tracking and Status Visibility:
Applicants have limited ability to monitor the progress of their submissions, while administrators face difficulties tracking application history and workflow stages. This lack of visibility reduces transparency and contributes to delays.

Poor License Lifecycle Management:
Manual tracking of license issuance, validity periods, and expirations increases the likelihood of oversight errors. Without automated alerts or renewal mechanisms, both professionals and regulators face challenges in maintaining compliance.

Fragmented Data Storage and Record Management:
Scattered records maintained through spreadsheets, paper files, and disconnected tools result in data redundancy, inconsistencies, and difficulties in retrieving historical information.

Limited Reporting and Analytical Capabilities:
The absence of centralized digital records restricts the ability of RRA officers to generate accurate reports, monitor trends, evaluate regulatory performance, and support data-driven decision-making.

These challenges collectively reduce operational efficiency, increase administrative workload, and negatively affect service delivery to applicants. The lack of automation and centralized data management also limits transparency, accountability, and effective regulatory oversight.

There is therefore a critical need to develop and implement a centralized digital system capable of automating tax professional management processes. Such a system should streamline application workflows, enable secure document submission and validation, support license lifecycle tracking, enhance transparency, and provide comprehensive reporting tools. By addressing these limitations, the proposed TPMS will improve efficiency, ensure data accuracy, strengthen regulatory control, and enhance the overall effectiveness of tax professional management at RRA.

Choice and Motivation of the Study

This study was undertaken in response to the need to enhance the efficiency and effectiveness of tax professional management processes at the Rwanda Revenue Authority (RRA). The regulation of tax professionals, including registration, licensing, renewal, and monitoring, is a critical function that supports tax compliance and institutional service delivery. However, existing procedures often rely on manual and semi-automated practices, which introduce challenges related to workflow efficiency, document handling, data consistency, and transparency. The growing number of tax professionals and the increasing complexity of regulatory requirements necessitate the adoption of a centralized and automated digital solution. The proposed Tax Professional Management System (TPMS) seeks to address these operational limitations by streamlining regulatory processes and improving administrative performance.

From an academic perspective, this project represents a practical application of the knowledge and technical competencies acquired during the Bachelor of Science in Information Technology program at the Adventist University of Central Africa (AUCA). The

study integrates core areas of Information Technology, including system analysis and design, database management, secure application development, and workflow automation. It aligns with AUCA's mission of developing innovative professionals capable of applying technology to solve real-world organizational challenges. The project therefore serves as both an academic requirement and a demonstration of applied technological problem-solving within a government regulatory environment.

From an institutional perspective, TPMS contributes to improving regulatory efficiency, transparency, and data integrity at RRA. By digitizing and centralizing tax professional management processes, the system enhances operational effectiveness through automated workflows, secure document management, license lifecycle tracking, and reporting functionalities. The study also provides valuable professional development by strengthening competencies in system development, security implementation, and enterprise application design. Furthermore, the project aligns with Rwanda's Vision 2050 and the National Strategy for Transformation (NST2), which emphasize digital governance, modernization of public services, and institutional efficiency.

Objectives of the Study

General Objective

The general objective of this study is to design and implement a Tax Professional Management System (TPMS) that automates and streamlines the registration, licensing, renewal, and monitoring processes of tax professionals at the Rwanda Revenue Authority (RRA). The system aims to provide a centralized digital platform that improves workflow efficiency, enhances data accuracy, strengthens transparency, and supports effective regulatory oversight. The platform will facilitate online application submission, secure document management, automated approval workflows, license lifecycle tracking, and reporting functionalities.

Specific Objectives

- To develop a secure, web-based platform that enables tax professionals and tax assistants to submit applications and renewal requests electronically, reducing reliance on manual and paper-based procedures.

• To design an intuitive and user-friendly interface that ensures ease of use for applicants and administrative officers, promoting accessibility and efficient system interaction.

• To design and implement a robust PostgreSQL database to securely store and manage applicant records, professional details, licensing information, supporting documents, and workflow history.

• To implement a secure document upload and validation mechanism that ensures compliance with regulatory requirements and prevents incomplete or inconsistent submissions.

• To design and integrate automated approval and rejection workflows that support structured application review, decision tracking, and transparency in regulatory processes.

• To develop a license issuance, renewal, and expiry tracking mechanism that enables effective lifecycle management and automated compliance monitoring.

• To implement reporting and data export functionalities that support regulatory monitoring, performance evaluation, and data-driven decision-making.

• To ensure data security, privacy, and system reliability through the implementation of authentication, authorization, and access control mechanisms in accordance with information security best practices.

Scope of the Study

This study focuses on the design and implementation of a web-based Tax Professional Management System (TPMS) for the Rwanda Revenue Authority (RRA). The system is intended to digitize and streamline key regulatory processes related to tax professionals, including registration, application processing, licensing, renewal, document management, workflow tracking, and reporting. The platform is designed to improve administrative efficiency, enhance data accuracy, and strengthen transparency by replacing manual and fragmented procedures with a centralized digital solution.

The scope of the project is limited to the management and regulation of tax professionals and does not extend to core tax collection operations, financial transactions, payment processing,

or integration with external national platforms. The study specifically addresses challenges associated with manual application handling, document verification, approval workflows, and record management. While the system architecture is designed to support scalability and future integration, the present study emphasizes the automation of fundamental regulatory workflows within the TPMS domain.

Furthermore, this research concentrates on system design, functionality, performance, and usability evaluation within the operational context of RRA. System testing and validation focus on user interaction, workflow efficiency, data management, and security mechanisms. The study prioritizes the development of a reliable and secure application that supports regulatory oversight, improves service delivery, and enhances institutional productivity.

Finally, the scope of this study does not include advanced predictive analytics, inter-agency system integration, or extended compliance enforcement modules. Such features may be considered as future enhancements. The findings and prototype developed through this study are therefore intended to serve as a foundational framework for continued system refinement and potential expansion in support of RRA's digital transformation initiatives.

Methodology and Techniques Used in the Study

This study adopts a combination of research methods to analyze existing challenges in the management and regulation of tax professionals and to design the Tax Professional Management System (TPMS) for the Rwanda Revenue Authority (RRA). These methods were selected to ensure accurate data collection, reliable analysis, and practical system development. The following techniques were applied:

Observation

Observation is a qualitative method that involves directly monitoring real-world activities to understand how processes function in practice. This allows identification of inefficiencies and areas for improvement that may not appear in documentation alone.

In this study, direct observations were conducted within the RRA offices to monitor the current manual processes for tax professional registration, license issuance, renewal, and record management. Several challenges were identified, including reliance on paper-based

forms, delays in application processing, fragmented record-keeping, and difficulties in tracking application status. Observation validated documentation findings and provided insights for designing a more efficient and automated system.

Interview

Interviews are a qualitative research method used to collect detailed insights from key stakeholders regarding their experiences, challenges, and operational needs. This method is essential for understanding existing workflows and identifying system requirements.

In this study, interviews were conducted with relevant personnel involved in the management of tax professionals at the Rwanda Revenue Authority (RRA). The discussions provided insights into current manual processes, highlighting challenges such as application delays, document management difficulties, and data inconsistencies.

The interviews focused on understanding workflow structures, identifying system needs, and capturing expectations for an automated Tax Professional Management System. The findings contributed significantly to defining system functionalities and ensuring alignment with organizational requirements.

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Expected Results

The implementation of the Tax Professional Management System (TPMS) is expected to generate significant improvements in the registration, licensing, renewal, and regulatory management of tax professionals at the Rwanda Revenue Authority (RRA). The system is anticipated to enhance operational efficiency, data accuracy, transparency, and overall service delivery.

Upon successful implementation of the system, the following outcomes are expected:

Improved Application Processing Efficiency:
The digital platform will streamline application submission, review, and approval workflows, reducing delays associated with manual procedures.

Centralized and Secure Data Management:
All tax professional records, licensing information, and supporting documents will be securely stored in a centralized database, improving data accessibility, consistency, and reliability.

Enhanced Document Management and Verification:
The system will enable structured document uploads and validation, minimizing errors, incomplete submissions, and document misplacement.

Effective License Lifecycle Management:
Automated tracking of license issuance, renewal, and expiry will improve compliance monitoring and reduce the risk of expired licenses.

Improved Communication and Status Tracking:
Applicants and regulatory officers will benefit from clearer visibility of application status and updates, reducing communication gaps and uncertainty.

Reduced Administrative Workload:
Automation of repetitive tasks such as data entry, record tracking, and reporting will increase staff productivity and operational efficiency.

Enhanced Reporting and Decision-Making:
The system will generate accurate reports and regulatory insights, supporting data-driven planning, monitoring, and policy formulation.

Strengthened Transparency and Accountability:
Structured workflows, digital records, and audit trails will improve process transparency and accountability in regulatory operations.

Contribution to Digital Transformation Initiatives:
By digitizing tax professional management processes, TPMS will support institutional modernization and align with broader digital governance objectives.

Organization of the Work

This research report is organized into five main chapters, each addressing a fundamental aspect of the study.

Chapter One:

Presents the general introduction, background of the study, statement of the problem, research objectives, scope of the study, research methodology, research questions, expected results, and overall structure of the report.

Chapter Two:

Examines the existing tax professional management processes at the Rwanda Revenue Authority (RRA), analyzes current operational procedures, and identifies key challenges and limitations associated with manual and semi-manual systems.

Chapter Three:

Describes the system requirements and design of the Tax Professional Management System (TPMS), including functional and non-functional requirements, system architecture, process models, and database design.

Chapter Four:

Details the system implementation and testing, outlining the development tools, technologies used, system functionalities, testing procedures, and challenges encountered during development.

Chapter Five:

Provides the summary of findings, conclusions drawn from the study, and recommendations for future enhancements and potential system expansion.

CHAPTER 2

ANALYSIS OF THE EXISTING SYSTEM

Introduction

The current system for managing tax professionals at the Rwanda Revenue Authority (RRA) is predominantly manual and involves fragmented administrative procedures. Core processes such as registration, application handling, document verification, licensing, renewal, and record management rely heavily on paper-based workflows. This approach leads to operational inefficiencies, processing delays, data inconsistencies, and increased administrative workload.

Manual application submission and document verification often result in delays, difficulties in tracking records, and increased risk of errors or data loss. The absence of a centralized digital platform limits transparency, making it challenging for applicants to monitor application status and for officers to efficiently manage regulatory workflows.

Communication between applicants and RRA officers is largely informal, relying on emails, phone calls, or physical visits. This creates communication gaps, delays, and inconsistencies in information exchange. Additionally, the lack of automated workflow management reduces efficiency in approval processes, renewal tracking, and reporting.

These limitations conflict with Rwanda's digital transformation objectives outlined in Vision 2050 and NST2, which emphasize efficiency, transparency, and technology-driven service delivery.

To address these challenges, the proposed Tax Professional Management System (TPMS) introduces a centralized web-based solution designed to automate registration, licensing, renewal, document management, workflow tracking, and reporting. The system aims to improve efficiency, enhance data accuracy, strengthen transparency, and support effective regulatory oversight.

Description of the Current System Environment

Historical Background

The Rwanda Revenue Authority (RRA) was established in 1997 by the Government of Rwanda to improve domestic revenue mobilization and strengthen tax administration. Its primary mandate is to assess, collect, and account for taxes and customs duties, while promoting voluntary compliance and providing efficient taxpayer services. Over the years, RRA has played a central role in Rwanda's economic development by modernizing tax systems and enhancing administrative efficiency through technology-driven reforms.

As part of its regulatory responsibilities, the RRA oversees the registration, licensing, and monitoring of tax professionals. Tax professionals serve as key intermediaries between taxpayers and the tax administration, providing advisory, compliance, and representation services. However, despite RRA's progress in digitalizing core tax operations, the management of tax professional registration, licensing, renewal, and document verification remains largely manual and fragmented.

The existing environment relies heavily on paper-based applications, manual document submission, and physical verification processes. These procedures often lead to processing delays, record management challenges, and difficulties in tracking application status. Additionally, communication between applicants and RRA officers is typically conducted through emails, phone calls, or physical visits, which may result in inefficiencies and inconsistencies.

Record-keeping within the current system presents further challenges, as tax professional data is stored across multiple administrative records or physical files. This limits real-time access to information, complicates reporting, and increases the risk of data inconsistencies or duplication. Moreover, the absence of a centralized digital platform restricts workflow automation, regulatory monitoring, and performance tracking.

In alignment with Rwanda's Vision 2050 and the National Strategy for Transformation (NST2), which emphasize digital governance and efficient public service delivery, there is a growing need for a modernized system to manage tax professional regulatory processes. The proposed Tax Professional Management System (TPMS) seeks to address these limitations

by introducing a centralized, web-based platform that automates registration, licensing, renewal, document management, workflow tracking, and reporting.

Mission

To enhance the efficiency, accuracy, and transparency of tax professional regulation by providing a secure, centralized, and user-friendly digital platform that streamlines registration, licensing, renewal, and regulatory oversight processes.

Vision

To establish a fully digital, integrated, and intelligent tax professional management framework that supports effective regulatory administration, improves service delivery, and aligns with Rwanda's digital transformation objectives.

Description of the Current System

The current system for managing tax professionals at the Rwanda Revenue Authority (RRA) operates largely through manual and paper-based procedures. Key processes such as tax professional registration, application submission, document verification, licensing, and renewal rely heavily on physical documentation and administrative handling.

Applicants seeking registration or licensing as tax professionals are required to submit paper-based forms along with supporting documents. These documents are reviewed and verified manually by authorized officers. Communication regarding application status, missing requirements, or approvals is primarily conducted through emails, phone calls, or physical visits to RRA offices.

This manual approach presents several challenges, including delays in processing applications, difficulties in tracking submitted documents, and risks of data inconsistencies. Record-keeping is often fragmented, with information stored across physical files or separate administrative records. This limits efficient access to applicant data and complicates regulatory monitoring and reporting activities.

Furthermore, the absence of a centralized digital platform makes it difficult to provide real-time visibility into application progress, license validity, and renewal deadlines. Officers

must rely on manual tracking mechanisms, which increases administrative workload and the likelihood of errors. Reporting processes are also time-consuming, as regulatory data must be compiled manually from multiple sources.

Overall, the existing system lacks automation, centralized data management, and real-time workflow tracking, which affects operational efficiency, transparency, and service delivery.

Key Issues with the Current System

The current system for managing tax professionals at the Rwanda Revenue Authority (RRA) faces several critical challenges that affect efficiency, accuracy, and transparency.

Manual Application and Registration Processes:
Tax professionals and applicants are required to submit paper-based forms and supporting documents physically. This manual approach increases processing time, creates administrative bottlenecks, and adds to the workload of RRA officers. It also limits convenience for applicants, particularly those located far from RRA offices.

Delays in Application Processing:
Application review, document verification, and approval procedures are conducted manually. As a result, applicants often experience delays in receiving feedback, approvals, or license issuance. The absence of automated workflows slows down decision-making and regulatory processes.

Lack of Real-Time Application Tracking:
There is no centralized mechanism that allows applicants or officers to monitor application status in real time. Applicants must rely on emails, phone calls, or physical visits for updates, which reduces transparency and increases uncertainty.

Inefficient Document and Record Management:
Tax professional records and supporting documents are primarily stored in physical files or fragmented systems. This increases the risk of document loss, duplication, data inconsistencies, and difficulties in retrieving information when needed.

Unstructured Communication Channels:
Communication between applicants and RRA officers is largely informal, relying on emails, phone calls, or in-person interactions. This may result in delayed responses, incomplete submissions, and miscommunication.

Absence of Automated Workflow Management:

Processes such as approval routing, renewal tracking, and license validation require substantial manual intervention. This reduces operational efficiency and increases the likelihood of administrative errors.

Limited Reporting and Monitoring Capabilities:

Regulatory reports are compiled manually, making the process time-consuming and prone to inaccuracies. The lack of centralized reporting tools limits effective monitoring, analysis, and evidence-based decision-making.

Analysis of the Current System

The current system for managing tax professionals at the Rwanda Revenue Authority (RRA) is largely manual and paper-based. Processes such as registration, application handling, document verification, licensing, and renewal rely on physical documentation and fragmented record-keeping. This approach results in processing delays, data inconsistencies, and difficulties in tracking application status.

The absence of a centralized digital platform limits transparency, as applicants cannot monitor progress in real time. Additionally, manual document management increases the risk of data loss, duplication, and administrative errors. Overall, the system lacks automation, efficiency, and effective workflow tracking, which affects regulatory oversight and service delivery.

Analysis of the Current System Based on Key Issues

An in-depth analysis of the current tax professional management processes at the Rwanda Revenue Authority (RRA) indicates that reliance on manual procedures significantly affects operational efficiency, accuracy, and transparency. Paper-based workflows and fragmented record management increase the risk of human error, data inconsistencies, and administrative delays. As the volume of applications grows, managing records, verifying documents, and tracking regulatory processes becomes increasingly complex and inefficient.

1. **Manual Application and Registration Processes**:

The current system depends on paper-based submission of applications and supporting documents. This manual approach increases administrative workload, slows processing times,

and exposes records to errors, duplication, and loss. As application volumes increase, the system becomes difficult to manage effectively.

2. **Delays in Application Processing and Approval:**
Document verification, evaluation, and approval workflows are handled manually, resulting in prolonged processing periods. Applicants often experience delays in receiving feedback or license issuance, which affects service delivery and user satisfaction.

3. **Lack of Real-Time Tracking and Transparency**:
The absence of a centralized digital platform prevents applicants and officers from tracking application status in real time. This limits transparency and forces reliance on informal communication channels such as emails or phone calls.

4. **Inefficient Document and Record Management:**
Tax professional data and supporting documents are stored across physical files or disconnected records. This fragmented storage increases the risk of data loss, duplication, and retrieval difficulties, while also complicating regulatory monitoring.

5. **Unstructured Communication Mechanisms**:
Communication between applicants and RRA officers is largely informal and reactive. This may result in delayed responses, incomplete submissions, and miscommunication regarding application requirements or status.

6. **Absence of Automated Workflow Management**:
Processes such as approval routing, renewal tracking, and license validation require substantial manual intervention. This reduces efficiency, increases administrative burden, and heightens the likelihood of errors.

7. **Limited Reporting and Analytical Capabilities**:
Regulatory reports are compiled manually, making them time-consuming and prone to inaccuracies. The lack of centralized data limits effective monitoring, performance evaluation, and evidence-based decision-making.

Overall, these challenges highlight the need for a centralized, automated, and secure digital solution. The proposed Tax Professional Management System (TPMS) is designed to address

these limitations by improving efficiency, enhancing transparency, strengthening data accuracy, and supporting effective regulatory oversight.

Modeling of the Current System

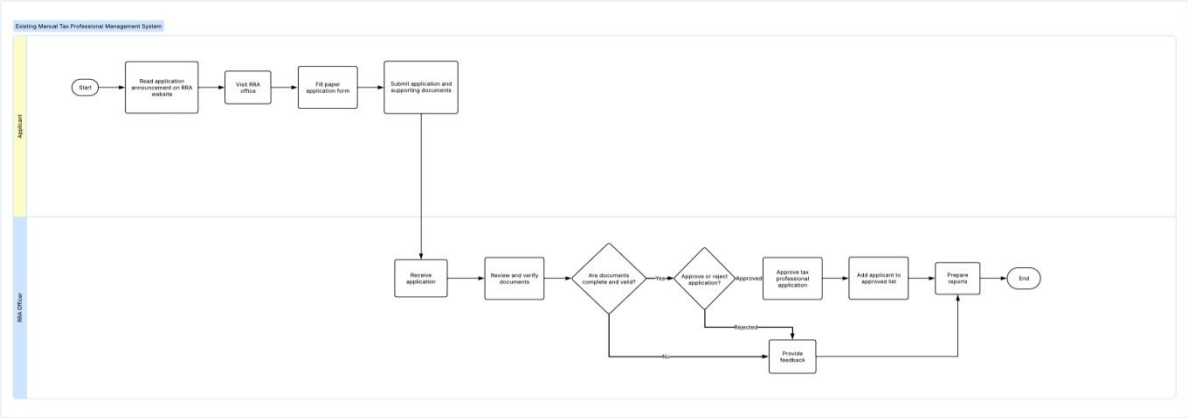


Figure 1: Current System Model

Here is an overview of the current tax professional applications at the Rwanda Revenue Authority (RRA) :

Step 1: Application Announcement:

The process begins when RRA publishes a notice on its website informing the public that applications for tax professionals are open. This announcement provides details about the requirements and eligibility criteria.

Step 2: In-Person Application Submission:

Interested applicants must visit RRA offices to submit their applications. The application is completed using paper-based forms, as there is no online submission system.

Step 3: Document Submission:

Along with the application form, applicants submit the required supporting documents. These typically include academic certificates, tax clearance certificates, identification documents, and other relevant records.

Step 4: Document Review and Verification:

RRA officers manually review the submitted documents to check whether the application is complete and meets the required conditions. If documents are missing or unclear, applicants are contacted for clarification or resubmission.

Step 5: Approval or Rejection:

After verification, authorized officers decide whether to approve or reject the application based on the established criteria.

Step 6: Reporting of Results:

Finally, RRA prepares reports listing approved and rejected applicants. These reports are compiled manually for administrative and regulatory purposes.

Problems of the Current System

Performance:

Throughput

- ❖ The current system relies on manual procedures for handling tax professional applications. Each application passes through several stages, including physical submission, document checking, verification, and approval. These repeated manual steps slow down the entire process.
- ❖ Because there is no automation, officers spend significant time performing routine tasks such as reviewing paper documents, organizing files, and updating records. As the number of applications increases, processing becomes slower and more difficult to manage.
- ❖ The use of paper-based workflows also increases the risk of document duplication, missing files, and inconsistent applicant information.

Response Time

- ❖ Communication within the current system is often delayed. Applicants do not receive immediate updates about their application status and must rely on emails, phone calls, or office visits.
- ❖ Document verification and approval decisions take longer due to manual review procedures. When documents are missing or incorrect, additional time is required to contact applicants and request corrections.
- ❖ The lack of real-time tracking tools makes it difficult for both applicants and officers to monitor progress efficiently.

Information

Input

❖ Applicant information is collected using paper-based forms without automated validation. This increases the likelihood of errors, incomplete submissions, and inconsistent data.

❖ Since there is no automatic data entry mechanism, officers must manually capture and verify applicant details, which is time-consuming and prone to mistakes.

❖ Updates or corrections to applicant information are not always recorded consistently, leading to outdated or conflicting records.

Output

❖ Application results, notifications, and updates are handled manually. Applicants may experience delays in receiving feedback on approvals, rejections, or required corrections.

❖ Reports are prepared manually, limiting accuracy and making it difficult to generate timely regulatory summaries.

❖ The absence of automated notifications reduces transparency and slows information sharing.

Storage

❖ Application forms and supporting documents are primarily stored in physical files or fragmented records.

❖ Paper-based storage creates risks such as document loss, damage, duplication, and difficulty retrieving historical records.

❖ The lack of a centralized digital database limits accessibility, backup mechanisms, and efficient record management.

Economics

❖ The manual system requires significant administrative effort, including printing forms, handling physical documents, and managing records.

❖ Officers spend considerable time on repetitive tasks such as document verification, filing, and report preparation, which increases operational costs.

❖ Delays and inefficiencies in the process lead to wasted resources and reduced productivity.

Control

❖ The current system lacks strong monitoring and tracking mechanisms.

❖ Manual processes make it difficult to consistently verify document authenticity, track application progress, and monitor license validity.

❖ The absence of automated controls increases the risk of errors, inconsistencies, and limited regulatory visibility.

Efficiency

❖ The system depends heavily on manual intervention at nearly every stage.

❖ Repeated manual handling of applications, documents, and records slows operations and increases the risk of administrative errors.

❖ Without automation and integration, workflow management remains inefficient and difficult to scale.

Service

❖ Applicants have limited visibility into their application status.

❖ There are no automated updates or real-time notifications, forcing applicants to rely on manual follow-ups.

❖ Delays and communication gaps reduce transparency and negatively affect user experience.

Proposed Solutions

To overcome the challenges identified in the current manual system, the study proposes the development of the Tax Professional Management System (TPMS). The system is designed to improve efficiency, accuracy, transparency, and service delivery.

1. **Centralized Digital Platform**

   The proposed system will provide a single digital platform where all tax professional applications and related activities are managed. Instead of relying on paper-based submissions, applicants will be able to submit their information and required documents electronically.

This centralized approach will simplify application handling, reduce administrative complexity, and improve record management.

2. **Improved Accessibility and User Experience**

   The TPMS will be designed with a user-friendly interface to ensure ease of use for both applicants and RRA officers. The system will be accessible online, allowing users to interact with the platform conveniently without the need for physical visits.

This will significantly reduce delays and improve overall user experience.

3. **Automated Application Processing**

   The system will automate key processes such as document validation, application review, approval workflows, and status updates. Automation will reduce repetitive manual work, minimize errors, and accelerate processing time.

Applicants will also receive timely updates regarding their application status.

4. **Enhanced Communication and Notifications**

   The TPMS will include notification mechanisms that keep applicants informed about approvals, rejections, missing documents, or required corrections. This will eliminate communication delays common in the manual system.

Clear and timely communication will improve transparency and user satisfaction.

5. **Secure Digital Data Management**

   All applicant information and documents will be stored in a centralized digital database. This will reduce risks associated with paper-based storage, such as document loss, duplication, or damage.

Digital storage will also improve data retrieval, backup, and security.

6. **Improved Monitoring and Control**

   The system will provide better tracking of applications, approvals, and license validity. Officers will be able to monitor processes more effectively, ensuring consistency, accountability, and regulatory compliance.

This will strengthen operational control and decision-making.

System Requirement

Functional Requirements

- **REQ 1:**
  The system shall allow tax professionals to **create an account and submit applications online** by providing personal information and required details.
- **REQ 2:**
  The system shall allow applicants to **upload required supporting documents** such as academic certificates, tax clearance certificates, and identification documents.
- **REQ 3:**
  The system shall enable RRA officers to **review and verify submitted applications and documents** through the system.
- **REQ 4:**
  The system shall allow RRA officers to **approve or reject tax professional applications** based on verification results.
- **REQ 5:**
  The system shall provide **notifications to applicants** informing them about the status of their applications (approved, rejected, or requiring additional information).
- **REQ 6:**
  The system shall allow RRA officers to **request additional documents or clarification** from applicants when necessary.
- **REQ 7:**
  The system shall enable administrators to **generate reports on tax professional applications**, including approved and rejected applications.
- **REQ 8:**
  The system shall implement **role-based access control**, allowing different system access levels for applicants, RRA officers, and administrators.

Non-Functional Requirements

- **NFR 1: Usability**
  The system shall provide a simple and user-friendly interface that allows applicants and

RRA officers to easily navigate and perform tasks such as submitting applications and reviewing documents.

- **NFR 2: Performance**

  The system shall process user requests and display information within a reasonable response time to ensure efficient operation.

- **NFR 3: Security**

  The system shall ensure secure access to data by requiring user authentication and protecting sensitive information such as applicant details and uploaded documents.

- **NFR 4: Reliability**

  The system shall operate consistently without frequent failures, ensuring that application data is stored safely and can be retrieved whenever needed.

- **NFR 5: Availability**

  The system shall be accessible online at all times, allowing applicants to submit applications and RRA officers to review them when required.

- **NFR 6: Data Integrity**

  The system shall maintain accurate and consistent data, ensuring that submitted applications and documents are not altered or lost during processing.

- **NFR 7: Scalability**

  The system shall be able to handle an increasing number of users and applications without reducing system performance.

- **NFR 8: Maintainability**

  The system shall be designed in a way that allows easy updates, maintenance, and improvements when needed.

CHAPTER 3

REQUIREMENTS ANALYSIS AND DESIGN OF THE NEW SYSTEM

Introduction

This chapter focuses on the analysis and design of the proposed Tax Professional Management System for the Rwanda Revenue Authority (RRA). The aim of the system is to improve the current manual process used for managing tax professional applications. In the existing system, applicants must physically submit application forms and supporting documents at RRA offices, which can lead to delays, paperwork, and difficulties in managing records.

The proposed system introduces a digital platform that allows applicants to submit their applications online while enabling RRA officers to review, verify, and manage applications more efficiently. To clearly represent the system structure and interactions, Unified Modeling Language (UML) diagrams based on Object-Oriented Methodology (OOM) are used. These diagrams help illustrate how users interact with the system and how the different components of the system work together.

Analysis and Design Methodology

The analysis and design methodology provides a structured approach for understanding the requirements of the proposed Tax Professional Management System and designing a solution that meets user needs. System analysis focuses on identifying the challenges in the current manual process, such as delays in application processing, difficulties in tracking submitted documents, and inefficiencies in managing application records.

System design defines the structure and components required to support the new system. This includes identifying system modules, user interactions, and data structures that will allow applications to be submitted, reviewed, and managed digitally. The design also emphasizes security, data accuracy, and ease of use to ensure that both applicants and RRA officers can use the system effectively.

Object Oriented Methodology

Different methodologies can be used for system development, such as the Waterfall model, Procedural Modeling, and Prototyping. However, this project adopts the Object-Oriented Methodology (OOM) because it supports the development of systems that are modular, scalable, and easier to maintain. This approach is suitable for the Tax Professional Management System since it allows the system to be designed using objects that represent real-world entities involved in the application process.

Object-Oriented Methodology is appropriate for this project because the system involves different users such as applicants, RRA officers, and administrators, which can be represented as objects within the system. In addition, key processes such as application submission, document verification, approval or rejection of applications, and report generation can be organized into reusable classes and components. The use of Unified Modeling Language (UML) diagrams also helps visualize the structure of the system and the interactions between users and system components.

In Object-Oriented Methodology, objects are instances of classes and represent real-world entities within the system. For example, the system may include classes such as Applicant, Application, and Report. Each class contains attributes (data) and methods (functions) that define how the system behaves. The modular structure of OOM allows the system to be divided into manageable components such as the application management module, document verification module, and reporting module, making the system easier to develop and maintain.

Unified Modeling Language

Introduction to UML

The Unified Modeling Language (UML) is a widely used modeling language in software engineering that helps developers design and document information systems. It provides a clear way to visualize how a system is structured, how its components interact, and how users communicate with the system. UML is therefore useful in representing the design of the proposed Tax Professional Management System for the Rwanda Revenue Authority (RRA).

UML was developed in the 1990s by James Rumbaugh, Grady Booch, and Ivar Jacobson, and later standardized by the Object Management Group in 1997. Since then, UML has become an international standard used to model different aspects of software systems, from simple applications to large information systems.

In this project, UML is used to represent important elements of the system such as the system actors, system processes, and interactions between users and the system. Through diagrams such as use case diagrams, activity diagrams, and class diagrams, UML helps illustrate how applicants submit applications, how RRA officers review them, and how the system manages and stores application data.

Design of the New System

System design provides the structure for how the proposed Tax Professional Management System will operate. It defines the system architecture, modules, interfaces, workflows, and data components that support the digital management of tax professional applications at the Rwanda Revenue Authority (RRA). The goal of the design is to create a system that improves efficiency, reduces manual work, and simplifies the application management process.

The proposed system focuses on developing an integrated digital platform that allows applicants to submit their applications and supporting documents online while enabling RRA officers to review, verify, and manage applications more efficiently. The system will also support key functions such as application tracking, approval or rejection of applications, and report generation.

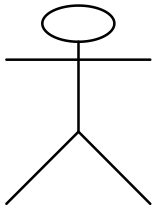
The following sections describe the system requirements, use cases, UML diagrams, and system models that guide the development of the new system. These models illustrate how the system will function, how users interact with it, and how application data is processed and managed within the system. Overall, the system is designed to improve transparency, accuracy, and efficiency in the management of tax professional applications at the Rwanda Revenue Authority.

Use Case Diagram

Introduction to Use Case Diagram

A Use Case Diagram in Unified Modeling Language (UML) illustrates how users interact with a system and the functions the system provides. It identifies the system users, known as actors, and the actions they can perform, referred to as use cases. In the proposed Tax Professional Management System, the diagram shows how actors such as applicants, RRA officers, and administrators interact with the system to perform tasks like submitting applications, uploading documents, reviewing applications, and generating reports.

Use case tools description

An actor: An actor is a person or external system that interacts with the system to perform certain tasks or receive services. Actors represent the roles played by users when interacting with the system. In a Use Case Diagram, an actor is usually represented by a stick figure and is placed outside the system boundary. Actor Name: The actor name identifies the role of the user interacting with the system. It is written below or near the actor symbol and describes the type of user, such as Applicant, RRA Officer, or Administrator.	 Actor Name
Subject Boundary The system boundary represents the limits or scope of the system being modeled. It is usually shown as a rectangle that contains all the use cases of the system. The name of the system is displayed at the top or inside the boundary.	System Name
Include Relationship An include relationship shows that one use case always uses the functionality of another use case. This means that the included use case is a required part of the main use case. It is represented by a dashed arrow labeled «include» pointing from the main use case to the included use case.	←
Extend Relationship An extend relationship represents optional or additional behavior that may occur in a use case under certain conditions. It is represented by a dashed arrow labeled «extend» pointing from the extending use case to the base use case.	→
Generalization Relationship A generalization relationship shows that one use case or actor is a specialized version of another more general use case or actor. It represents inheritance of behavior and is illustrated with a solid arrow with a hollow triangle pointing to the more general element.	↑

Association Relationship An association relationship represents the interaction between an actor and a use case. It shows that the actor participates in or communicates with that use case. This relationship is usually represented by a simple straight line connecting the actor to the use case	

Table 1: Use case tools description

Use Case diagram

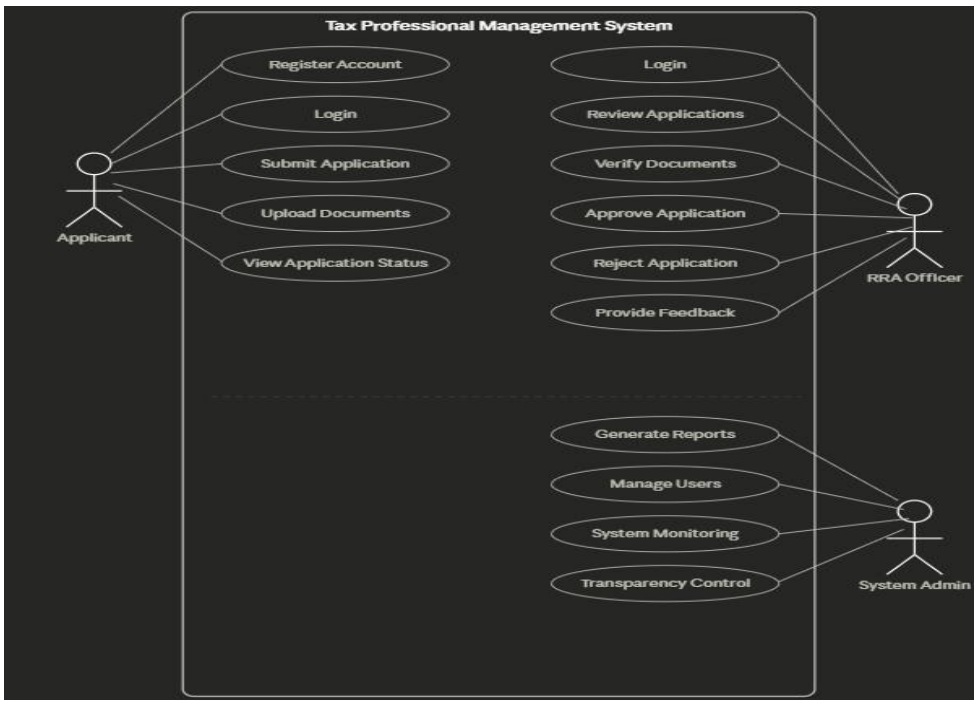


Figure 2: Use Case diagram

The Use Case Diagram shows how the main users interact with the proposed Tax Professional Management System. It illustrates the relationships between the system and its primary actors, such as applicants, RRA officers, and administrators, and the main functions they perform. These functions include submitting applications, uploading documents, reviewing applications, approving or rejecting applications, and generating reports. By presenting these interactions, the diagram provides a clear overview of the system's scope and how different users depend on the system to perform their tasks efficiently.

Class Diagram

Introduction to Class Diagram

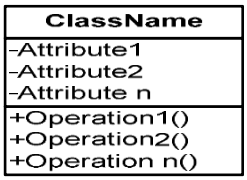
A Class Diagram is a type of UML diagram that shows the structure of a system by representing its classes and the relationships between them. It provides a static view of the system and helps in understanding how different parts of the system are organized.

Class diagrams describe the attributes (data), methods (functions), and relationships of the classes used in the system. They are important in object-oriented system design because they help developers model real-world entities within the system. In the Tax Professional Management System, the class diagram represents classes such as Applicant, Application, Document, and Report, and shows how these classes interact with each other within the system.

The elements of class diagram

Class

A class represents a real-world entity that the system needs to manage. In a class diagram, a class is shown as a rectangle divided into three parts: the class name at the top, attributes in the middle, and operations (methods) at the bottom. It represents objects such as Applicant, Application, or Document within the system.



Attribute

An attribute describes the properties or characteristics of a class. Attributes define the data stored in an object and represent its state. Derived attributes are indicated by placing a slash (/) before the attribute name.

Operation

An operation represents the functions or actions that a class can perform. Operations are written in the bottom section of the class and usually include parentheses that may contain parameters. They can represent actions such as creating, updating, or retrieving information.

Association

Association represents a relationship between two or more classes. It shows how classes are connected and interact with each other in the system. Associations may also include multiplicity symbols to indicate how many instances of a class are involved in the relationship.

Generalization

Generalization represents an “is-a” relationship between classes, where one class is a specialized version of another. It allows a class to inherit attributes and operations from a more general class.

Aggregation

Aggregation represents a “part-of” relationship between classes where one class contains or is made up of other classes. It is a special type of association that shows a logical grouping between related classes.

Composition

Composition is a stronger form of aggregation that represents a physical “part-of” relationship between classes. In this relationship, the existence of the contained object depends on the container object.

Schema of Class Diagram

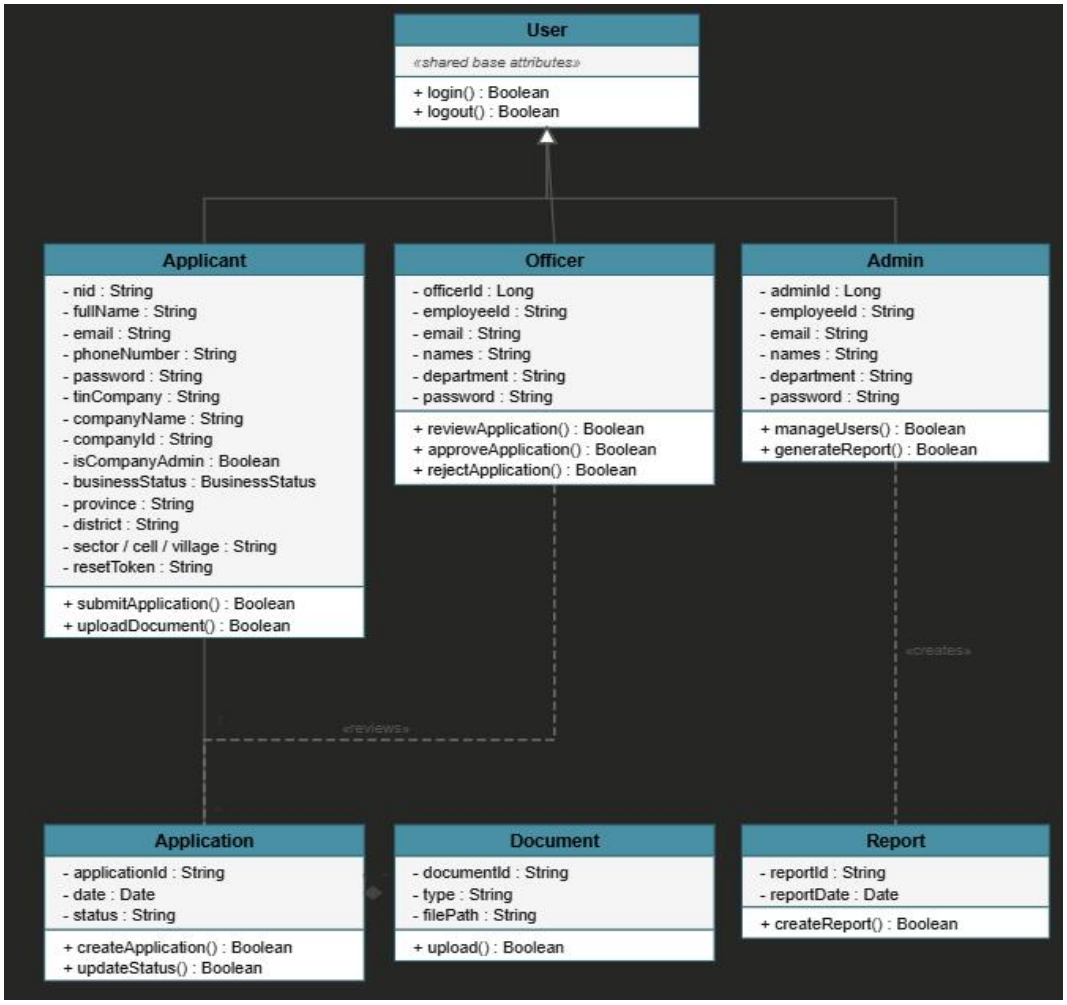


Figure 3: Schema of Class Diagram

Sequence Diagrams

Introduction to Sequence Diagrams

A Sequence Diagram is a UML interaction diagram that shows how users and system components communicate in a specific order while performing a task. It illustrates the sequence of messages exchanged between actors and the system, helping to understand the flow of activities during system operation. In the Tax Professional Management System, the sequence diagram shows how the applicant, RRA officer, and the system interact during processes such as application submission, document review, and approval or rejection of applications.

Applicant Submitting an Application

This sequence diagram illustrates how an applicant interacts with the Tax Professional Management System to submit an application. The applicant starts by logging into the system and filling in the application form, then uploads the required supporting documents. The system stores the application in the database and sends it to the RRA officer for review. The officer verifies the submitted information and decides whether to approve or reject the application. Finally, the system updates the application status and sends a notification to the applicant informing them of the result, ensuring a clear and organized application process.

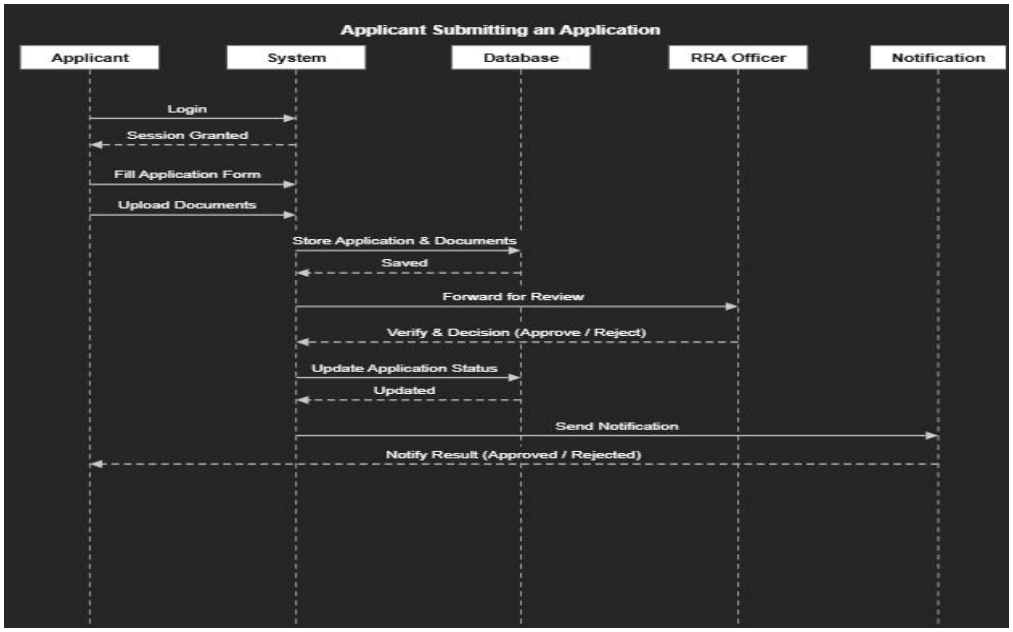


Figure 4: Applicant Submitting an Application

Officer Reviewing and Managing an Application

This sequence diagram shows how the RRA officer reviews and manages applications in the Tax Professional Management System. The officer logs into the system and selects an application submitted by the applicant. The system retrieves the application details from the database and displays them for verification. The officer checks the uploaded documents and decides whether to approve or reject the application.

If the officer updates the application status, the system saves the changes in the database and sends a confirmation message. The system then notifies the applicant about the decision, ensuring that all application information remains accurate and up to date while allowing officers to manage applications efficiently.

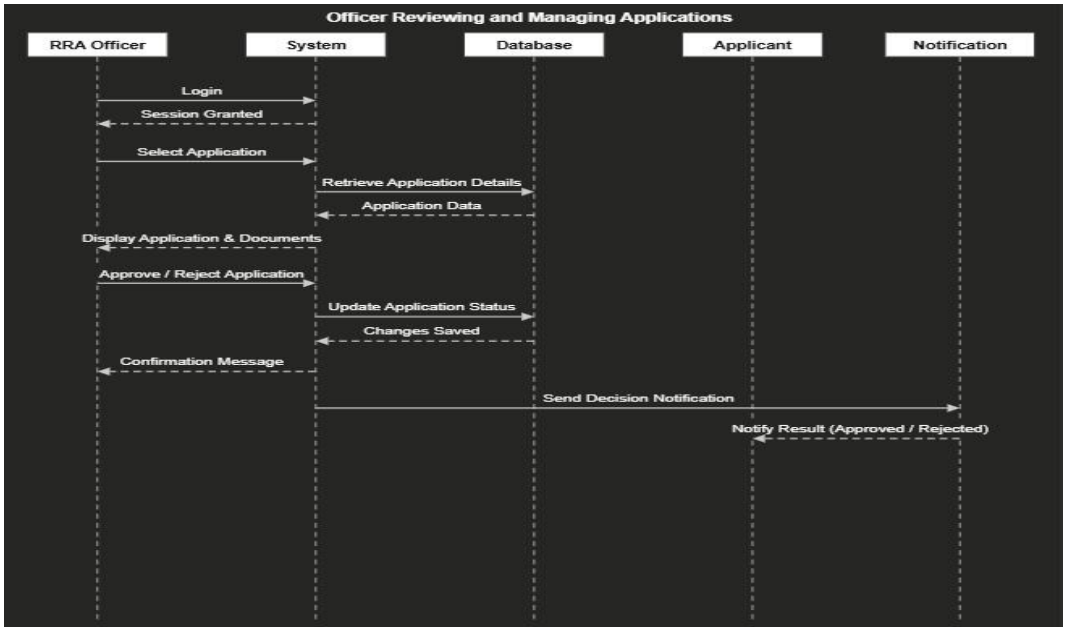


Figure 5: Officer Reviewing and Managing an Application

System Checking Application Status and Providing Decision

This sequence diagram shows how the system processes and checks an application after it has been submitted in the Tax Professional Management System. When an applicant requests to view the application status, the system sends a request to the database to retrieve the stored application details. The system then verifies the current status of the application, including whether it is pending, approved, or rejected.

If the application has already been reviewed by the RRA officer, the system returns the updated result and displays it to the applicant. This process ensures that applicants receive accurate and timely information about their applications, improving transparency and efficiency in the management of tax professional registration.

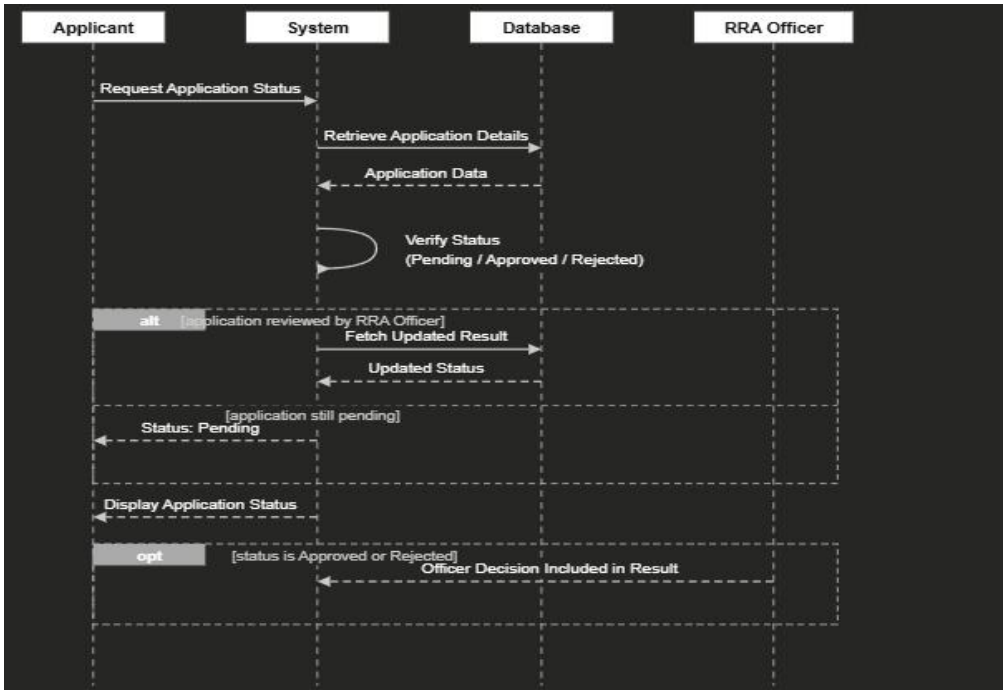


Figure 6: System Checking Application Status and Providing Decision

Activity Diagram

An activity diagram is a UML behavioral diagram that shows the flow of activities in a system, including the sequence of actions and decision points required to complete a task.

Applicant Submitting and Tracking Application

This activity diagram illustrates the process an applicant follows to submit and track an application in the Tax Professional Management System. The process begins when the applicant opens the system and logs into their account. The applicant fills in the application form and uploads the required documents, after which the system stores the information in the database.

At the decision point, the RRA officer reviews the application and decides whether to approve or reject it. If the application is approved, the system updates the status and issues the license. If it is rejected, feedback is provided to the applicant. Finally, the system sends a notification to the applicant showing the result, ensuring a clear and organized application process.

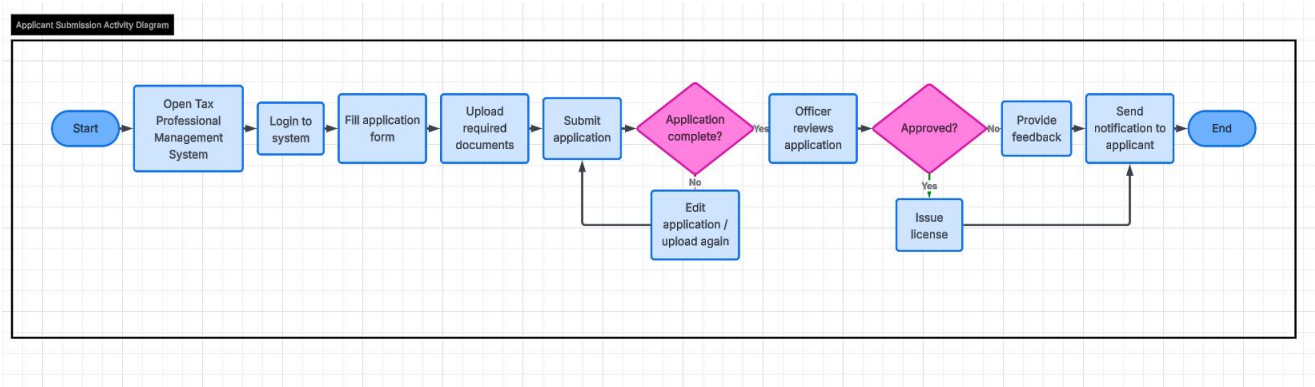


Figure 7: Application Verification Workflow

Officer Reviewing Applications

This diagram shows the process followed by the RRA officer in the Tax Professional Management System to review applications. The officer logs into the system, checks the submitted application and documents, and makes a decision to approve or reject it. The system then updates the application status and notifies the applicant, ensuring an accurate and well-organized application management process.

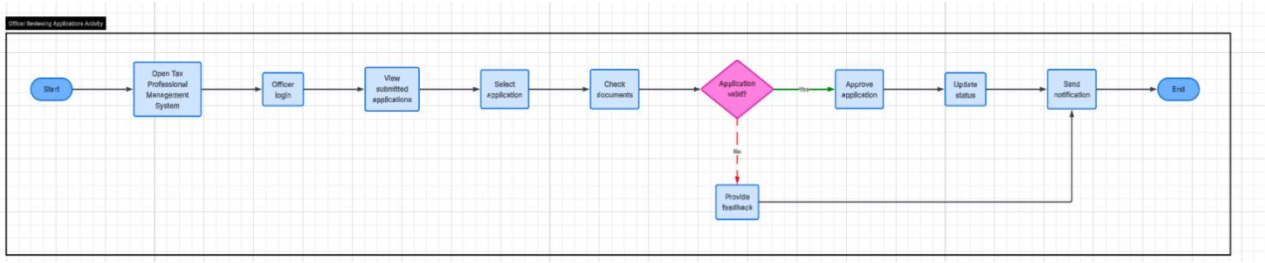


Figure 8: officer Reviewing Applications

Officer Managing Applications

The process starts when the officer logs into the system and reviews submitted applications stored in the database. The officer verifies the information and documents, then decides to approve or reject the application. The system saves the decision and updates the records, ensuring that application information remains accurate and properly managed.

Application Verification Workflow

This activity diagram shows the process the system follows to verify applications in the Tax Professional Management System. When an application is submitted, the system stores the applicant's information and documents in the database. The RRA officer reviews the

application, and the system checks the status to determine whether it should be approved or rejected. After the decision is made, the system updates the application status and displays the result to the applicant, ensuring an accurate and efficient application management process.

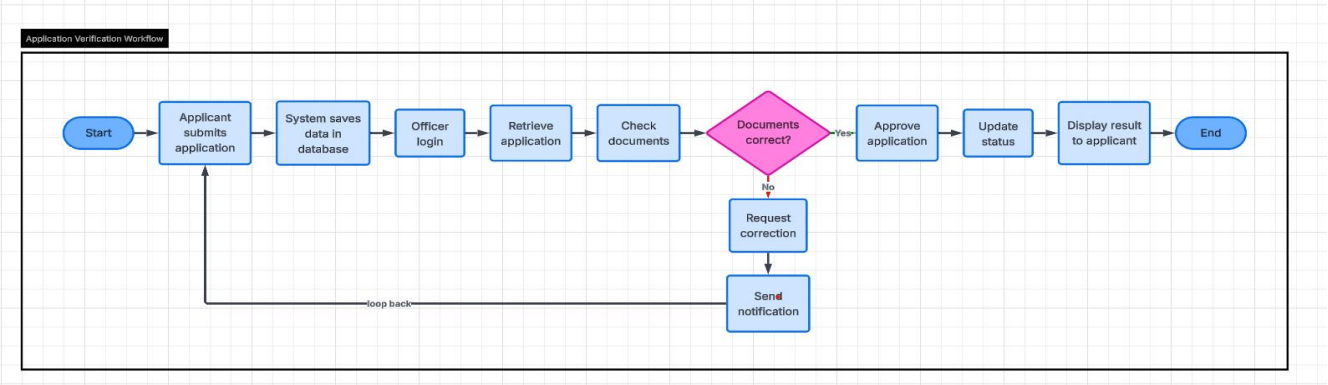


Figure 9: Application Verification Workflow

Database Schema Diagram

A database schema diagram shows how data is organized in the system and defines the structure of tables and their relationships. It ensures that data is stored accurately and consistently using rules provided by the database management system. In the Tax Professional Management System, the schema supports storing and managing data for applicants, applications, documents, officers, and reports while maintaining data integrity and good system performance.

Symbols for Database diagram

An entity is an object or concept about which the system stores information. In the database, an entity represents a real-world object such as Applicant, Application, Officer, or Document.	
Cardinality defines how many instances of one entity can be related to another entity. It shows relationships such as one-to-one, one-to-many, or many-to-many between entities.	
A connecting line is used to link entities and show the relationship between them in the diagram. It helps to represent how different tables in the database are related.	

An attribute is a property that describes an entity. Attributes represent the data stored in the database, such as name, ID, date, or status.	
A relationship shows the association between two or more entities. It explains how entities interact with each other within the database system.	

Figure 10: Symbols for Database diagram

Database Diagram Design

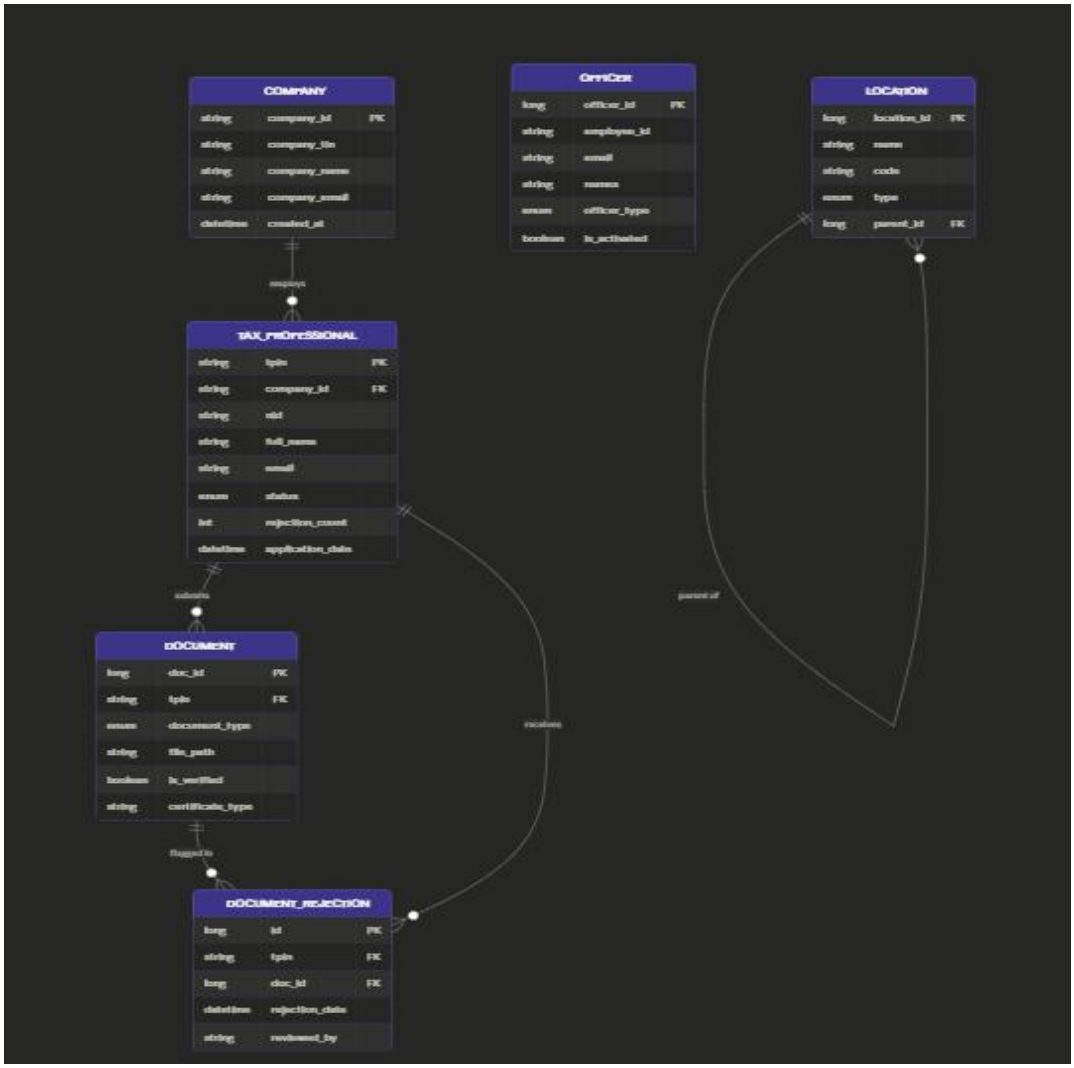


Figure 10: Database Diagram Design

Data Dictionary

A data dictionary describes the structure of data stored in the system and explains the attributes and relationships between entities. It helps developers and users understand how the database works and ensures data consistency during system development and maintenance.

TaxProfessional Table

This table stores basic information about applicants in the Tax Professional Management System, such as ID, name, and contact details, and is used to link applicants to their applications and documents.

Field Name	Data Type	Key	Description	Constraints / Notes
tpin	VARCHAR(50)	PK	Unique identifier for the volunteer.	Auto-generated/Required
tin_company	VARCHAR(50)		Full name of the volunteer.	Required
company_name	VARCHAR(50)		Name of the company.	Required
nid	VARCHAR(50)		National id of the a tax advisor.	Required
full_name	VARCHAR(50)		Full name of advisor.	Required
Email	VARCHAR(50)		Email address of the professional.	Required
phone_number	VARCHAR(50)		Primary contact phone number	Required
reset-token	VARCHAR(255)		Token used for password reset	Unique/optional

Reset_token_expiry	DATETIME		Expiry timestamp for the reset token	optional
province	VARCHAR(255)		Province of your workplace	Required
District	VARCHAR(255)		District of your workplace	Required
Sector	VARCHAR(255)		Sector of your workplace	Required
Cell	VARCHAR(255)		Cell of your workplace	Required
Sector	VARCHAR(255)		Sector of your workplace	Required
Village	VARCHAR(255)		Village of your workplace	Required
Business_status	Enum		Business status of the professional	Required
bachelor degree	Enum		Bachelor degree qualification	Required
Masters_degree	Enum		Masters degree qualification	Optional
Professional_qualification	Enum		Professional certification type	Required
Other_professional_details	VARCHAR(500)		Details for other certification type	optional

Other_professorial_file_past	VARCHAR(255)		File path for supporting qualification	optional
Application_date	DATETIME		Date the application was submitted	Auto-set on cerate
status	Enum		Current status of the application	Required(e.g PENDING , APPROVED, REJECTED)
Reviewed_by	VARCHAR(255)		NameID of officer who reviewed	Auto-set by the system
Reviewed_at	DATETIME		Timestamp of the review	Auto-set on reviewed date
Approval_date	DATETIME		Expiry data of the approval	Auto-set on approval date
Rejection_reason	VARCHAR(1000)		Reason for rejection	Required
Certification_file_path	VARCHAR(255)		File path to issued certifcate	Auto-sent by the system
Previous_reject_reason	VARCHAR(1000)		Archived reaction reason before	Required
Previous_reviewed_by	VARCHAR(255)		Officer who made the previous rejection	Auto-set by the system
Previous_reviewed_at	DATETIME		Timestamp of the previous rejection	Auto-set on reviewed date

rejection_count	INTEGER		Total number of times application was rejected	Default: 0
Reapplication_date	DATETIME		Date applicant reapplied after rejection	Auto-set on reapplication_date
Is_reapplication	BOOLEAN		Indicates if this is a reapplication	Default: false

Table 2: TaxProfessional Table

Company Table

Field Name	**Data Type**	**Key**	**Description**	**Constraints / Notes**
Company_id	VARCHAR(50)	PK	Unique identifier for the company.	Auto-generated/Required
Company_tin	VARCHAR(9)		Tax Identification Number of the company	Required/ Unique
Company_name	VARCHAR(255)		Official registered name of the company.	Required
Company_email	VARCHAR(255)		Primary contact email of the company.	Required/ Unique
Password	VARCHAR(255)		Hashed login password for company account.	Required
Reset_token	VARCHAR(255)		Token used for password reset.	Unique / Optional
Reset_token_expiry	DATETIME		Expiry timestamp for the reset token.	Optional

Province	VARCHAR		Province where the company is located	Required
Sector	VARCHAR		Sector where the company is located	Required
Cell	VARCHAR		Cell where the company is located	Required
Village	VARCHAR		Village where the company is located	Required
Created_at	VARCHAR		Timestamp where the company account was created	Auto-set on create

Table 3: Company Table

Document Table

Field Name	**Data Type**	**Key**	**Description**	**Constraints / Notes**
Doc_id	BIGINT	PK	Aut-generated unique identifier for the document.	Auto-generated/Required
tpin	VARCHAR(50)	FK	References the owning tax professional	Links to **TaxProfessional.tpin**
Document_type	ENUM		Type/category of the document	Required
File_path	VARCHAR(255)		Server file path to the	Required

			stored document	
Uploaded_at	DATETIME		Timestamp when the document was uploaded	Auto-set on create
is_verified	BOOLEAN		Indicates if the document has been verified	Default: false
Certificate_type	VARCHAR(50)		Sub-type for education certificates	Required
Bachelor_degree	ENUM		Bachelor's degree type	Required
Professional_qualification	ENUM		Professional qualification type	Optional
Other_professional_qualification	VARCHAR (500)		Details when qualification is OTHER	Optional
Masters_degree_name	VARCHAR (255)		Name of the master's degree if applicable	Optional

Table 4: Document Table

DocumentRejection Table

Field Name	**Data Type**	**Key**	**Description**	**Constraints / Notes**
id	BIGINT	PK	Aut-generated unique identifier.	Auto-generated/Required

tpin	VARCHAR(50)	FK	Reference taxProfessional whose document was rejected	Links to **TaxProfessional.tpin**
Doc_id	BIGINT	FK	Reference the rejected document	Links to Document.doc_id
rejection_date	DATETIME		Date and time when the document was rejected	Auto-set on create
Reviewed_by	VARCHAR(255)		Name of the officer who rejected the document	Required

Table 5: DocumentRejection Table

Officer Table

Field Name	**Data Type**	**Key**	**Description**	**Constraints / Notes**
Officer_id	BIGINT	PK	Auto-generated unique identifier for Officer.	Auto-generated/Required
employee_id	VARCHAR(100)		Employee id of the officer	Required / Unique
email	VARCHAR(255)		Officer's email	Required / Unique
names	VARCHAR(255)		Full names of Officer.	Required
department	VARCHAR(255)		Department the officer belong to	optional
password	VARCHAR(255)		Hashed login password	Optional (support initation flow)
Officer_type	ENUM		Role/type of the officer	Required (eg : Admin, officer)
Invitation_token	VARCHAR(255)		Token sent via invitation email	Unique / Optinal
Token_expiry	DATETIME		Expiry timestamp for the invitation token	Optional

Reset_token	VARCHAR(255)		Token used for password reset	Unique / Optional
Reset_token_expiry	DATETIME		Expiry timestamp for the reset token	Optional
Is_activated	BOOLEAN		Indicates if the officer has activated their account	Default:false
Crated_at	DATETIME		Timestamp when the officer account was created	Optional
Activated_at	DATETIME		Timestamp when the officer account was activated	optional

Table 6: Officer Table

Location table

Field Name	**Data Type**	**Key**	**Description**	**Constraints / Notes**
Location_id	BIGINT	PK	Aut-generated unique identifier for location.	Auto-generated/Required
name	VARCHAR(255)		Name of the Location	Required
code	VARCHAR(100)		Unique code identifier of the location	Required / unique
type	ENUM		Date and time when the document was rejected	Required (eg: province, District, sector, cell, village)
parent_id	BIGINT	FK	Name of the officer who rejected the document	Self-refential FK

Table 7: Location Table

System Architecture Design

System architecture design refers to the process of creating a blueprint or plan for how various components of a software system or application will interact with each other to achieve the desired functionality, performance, scalability, and reliability.

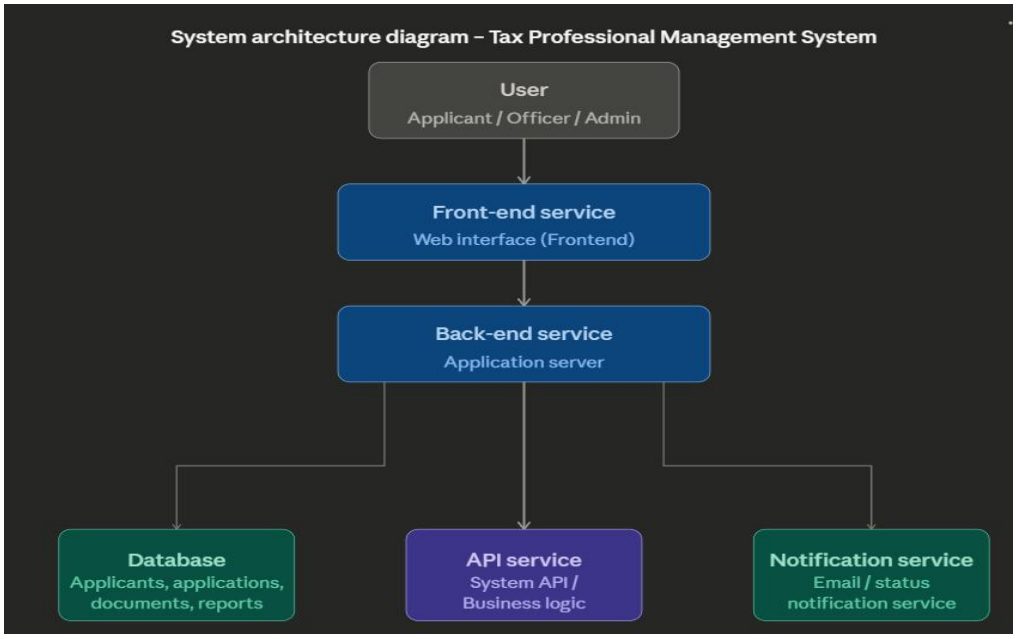


Figure 11: System Architecture Design

CHAPTER 4

IMPLEMENTATION OF THE NEW SYSTEM

Introduction

This chapter presents the implementation process of the Tax Professional Management System, a web-based platform developed to manage the registration, verification, and approval of tax professionals. The implementation phase focused on transforming the system design into a working application that improves the efficiency, accuracy, and transparency of application processing within the organization.

The system was developed using Object-Oriented Methodology together with an iterative development approach, allowing gradual improvement and testing of system components. The implementation involved setting up the development environment, creating the user interface, developing backend services, connecting the system to the database, and implementing features such as application submission, document verification, approval processing, and report generation. The final system provides a secure and well-structured platform for applicants, officers, and administrators to perform their tasks effectively.

Technologies Used

The Tax Professional Management System was developed using modern web technologies to ensure good performance, security, and scalability. The selected technology stack supports efficient application management, secure data handling, and smooth interaction between the system components.

Front End

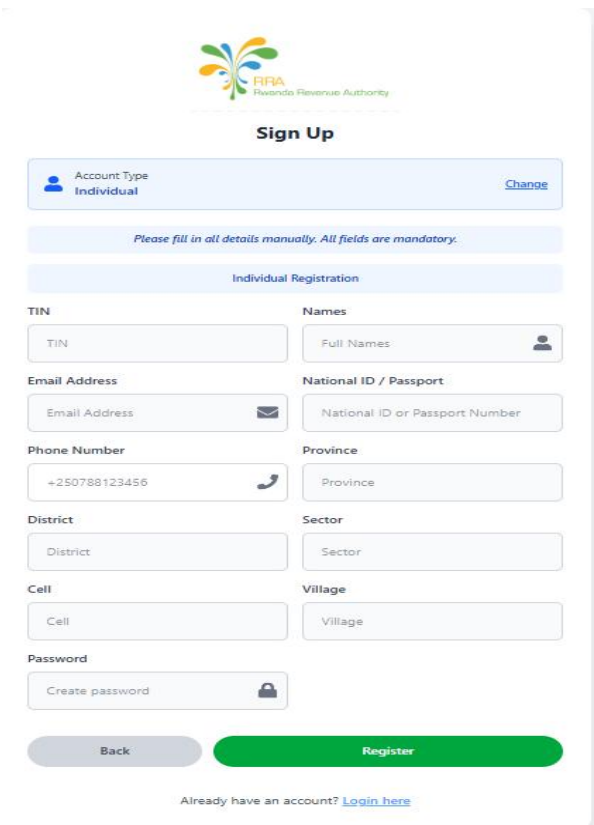
- **React with TypeScript (Vite)** – Used to build a fast and responsive user interface. The component-based structure allows easy development of pages such as application forms, dashboards, and management panels.
- **Tailwind CSS** – Provides a clean and responsive design, making the system easy to use on different screen sizes and improving the overall user experience.
- **Axios / Fetch API** – Used to send requests to the backend API and receive data such as applications, documents, and reports in real time.

Back End

- **Spring Boot (Java 17)** – Used as the main backend framework to implement the business logic of the system, including application processing, document verification, and report generation. It provides a structured and secure architecture for handling system operations.
- **Spring Security with JWT Authentication** – PostgreSQL – Used as the main database to store applicant information, applications, documents, and reports while maintaining data integrity and reliability.
- **PDF and Excel Processing Libraries** – Integrated into the backend to support generation and export of reports and official documents in PDF and Excel format.
- **PostgreSQL** – Used as the main database to store applicant information, applications, documents, and reports while maintaining data integrity and reliability.

Presentation of the New System

Applicants signup Page



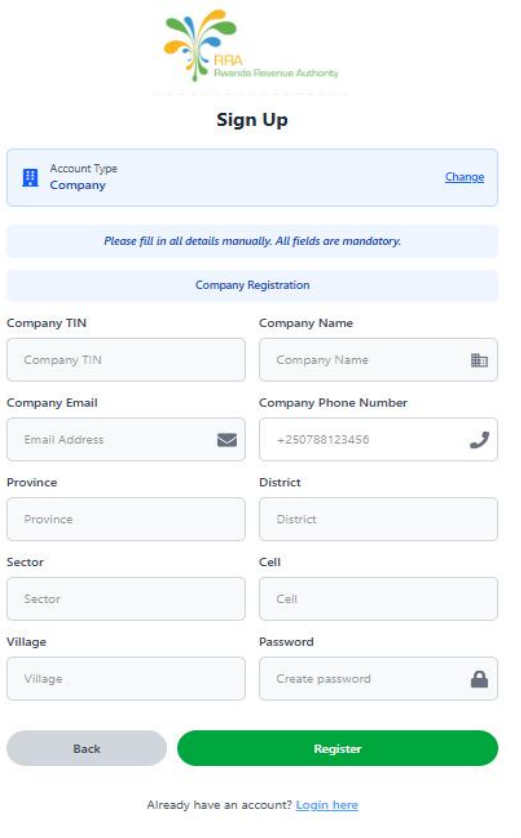


Figure 12 Applicant signup page

Login page

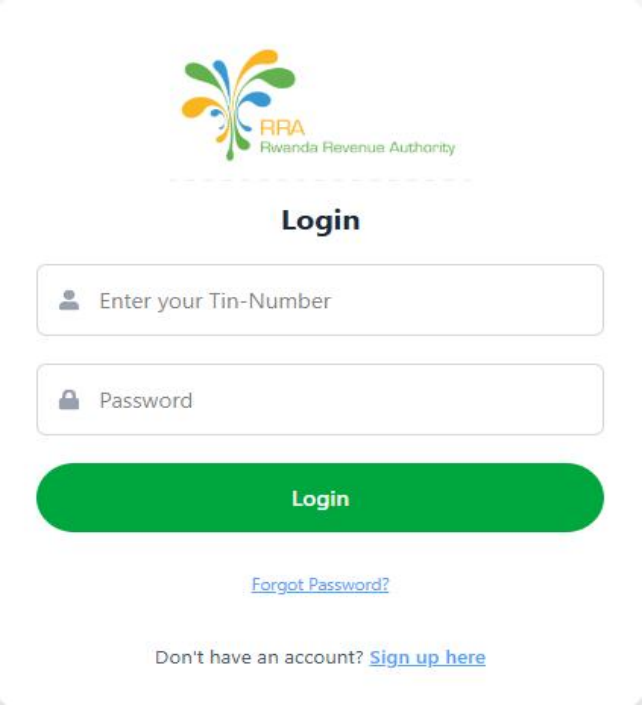


Figure 13 login page

Applicant Dashboard

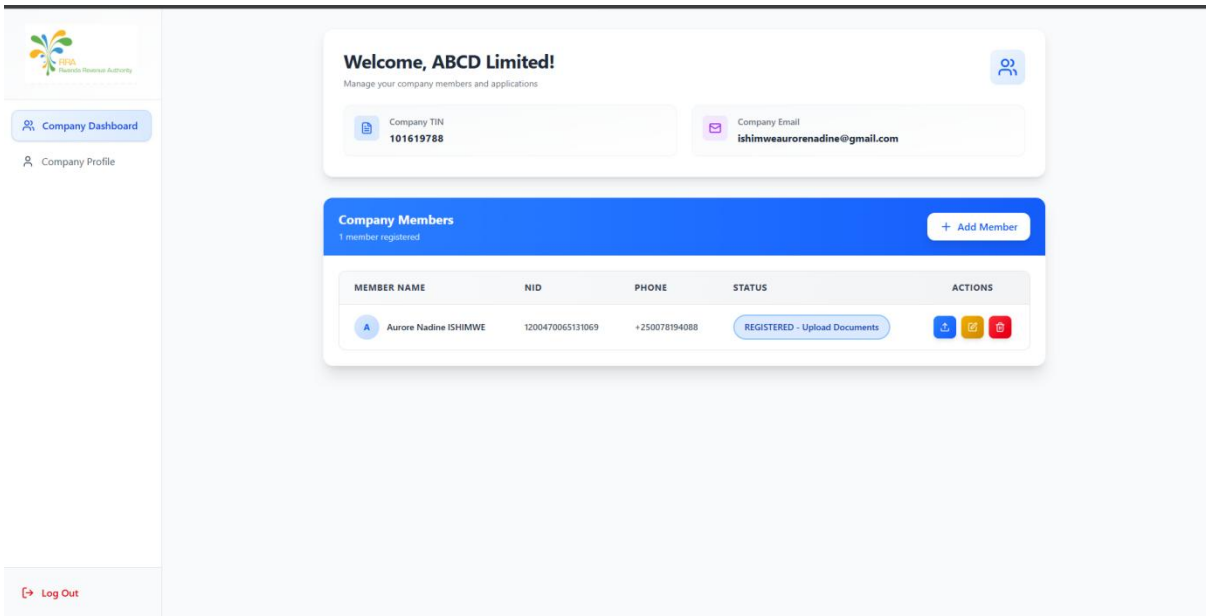


Figure 14: Applicant Dashboard

Document upload page

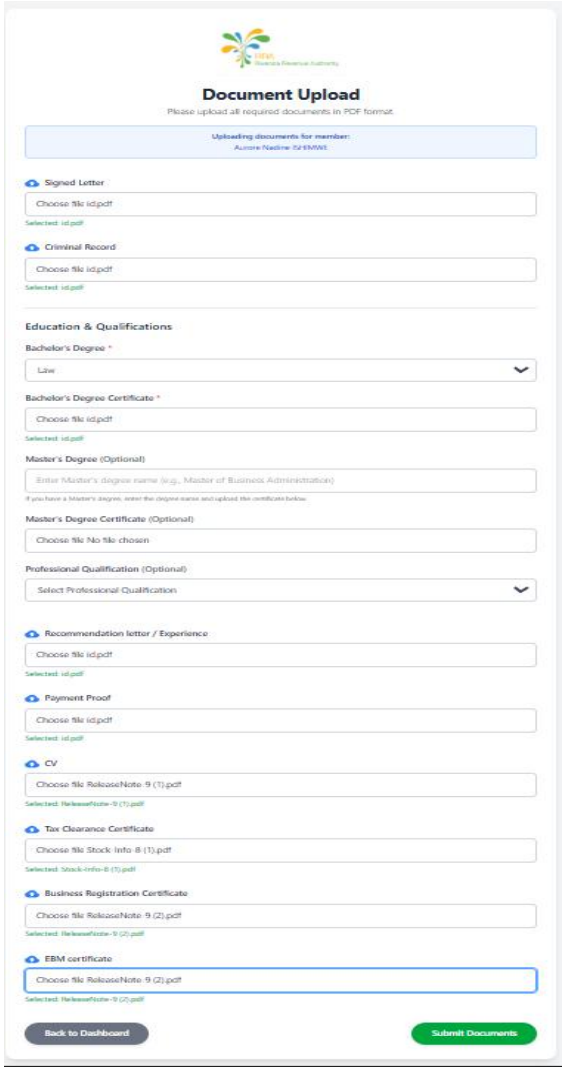


Figure 15: Document upload page

Company Profile

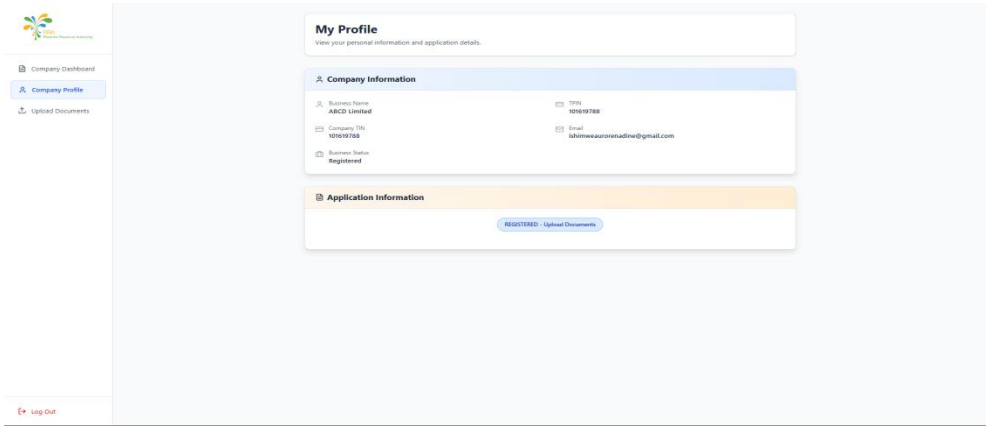


Figure 16: company profile page

Officers and Admin Login Page

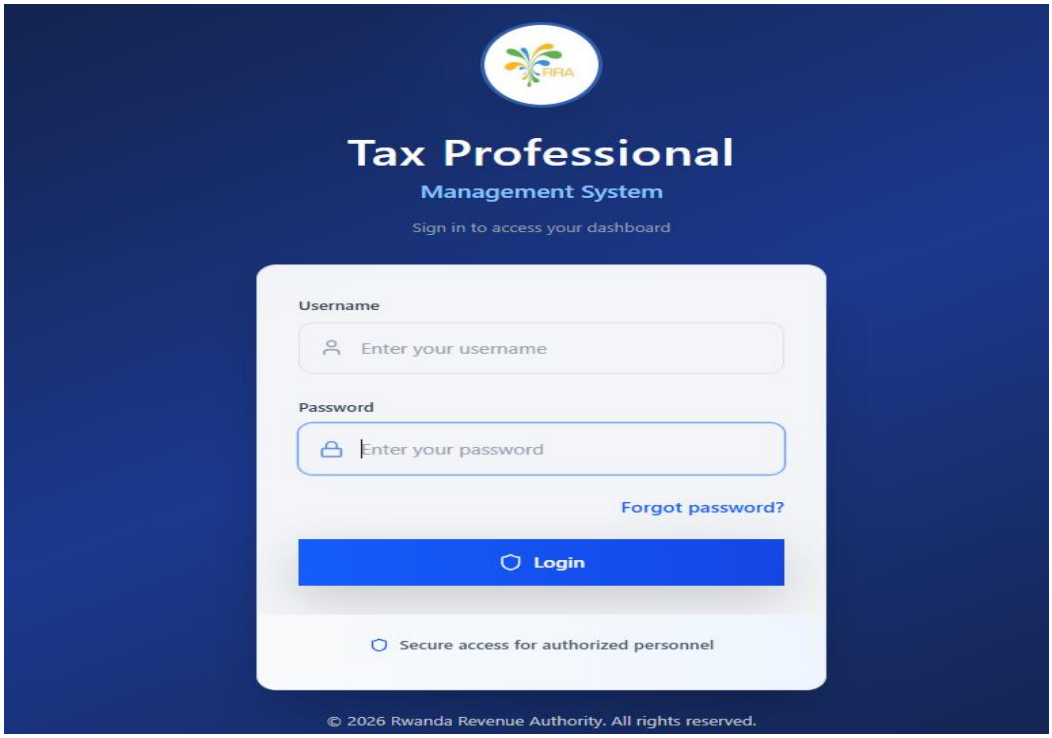


Figure 17: Officers and Admin Login page

Officers and Admin Dashboard Page

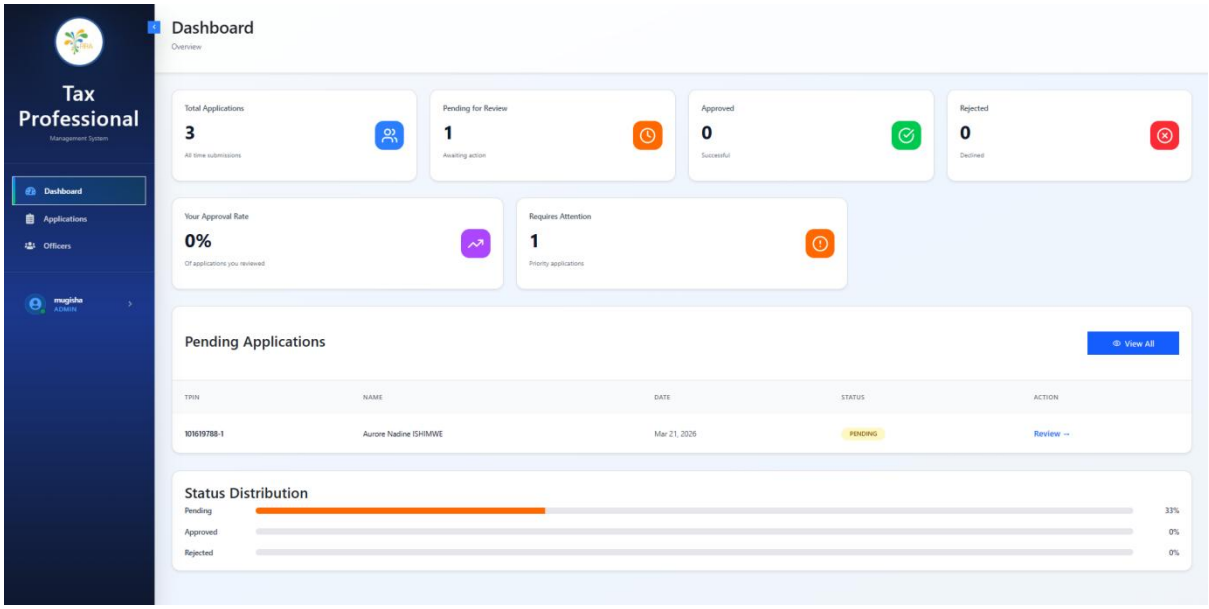


Figure 18: Officers and Admin Dashboard page

Officers Management Page

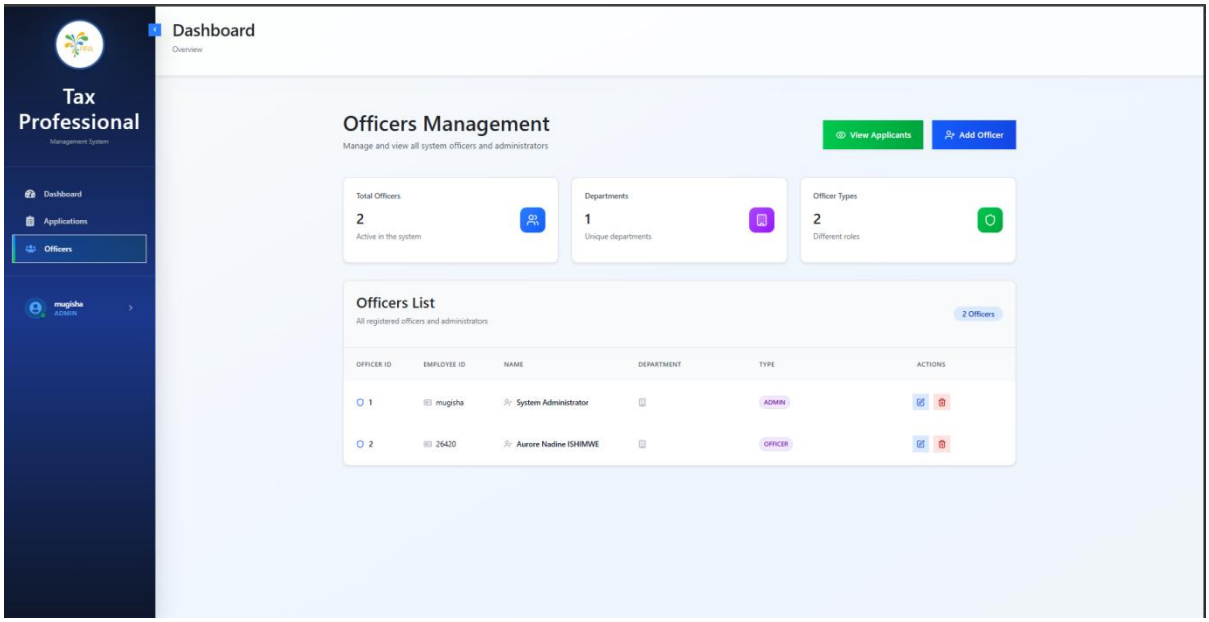


Figure 19: Officers Management page

Application Management Page

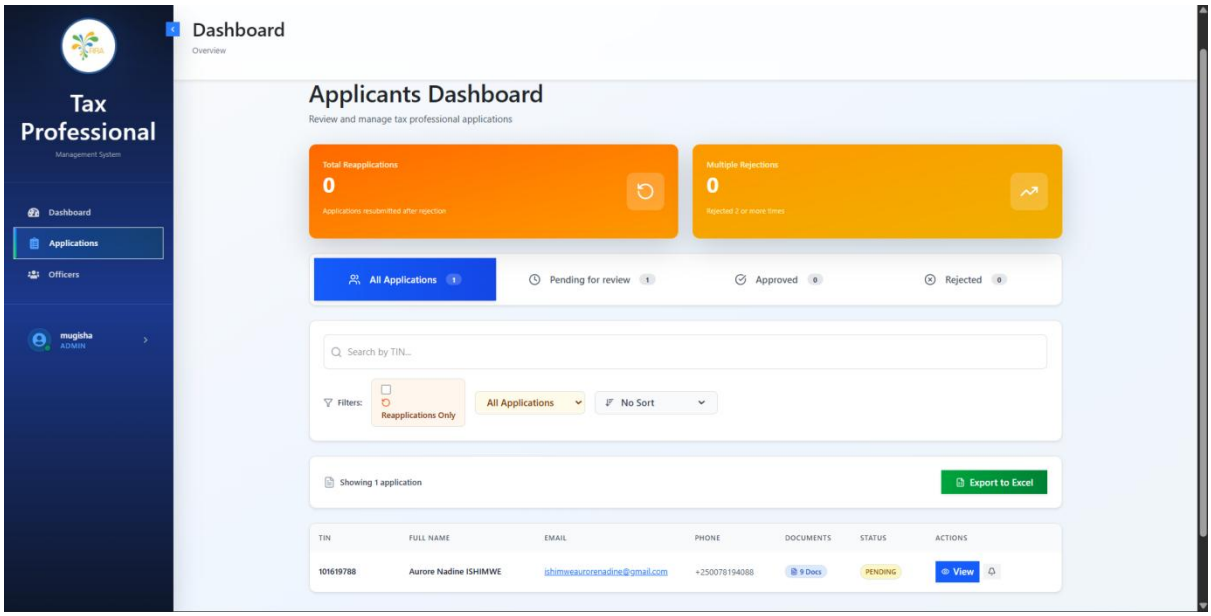


Figure 20: Application Management Page

Application Detail Page

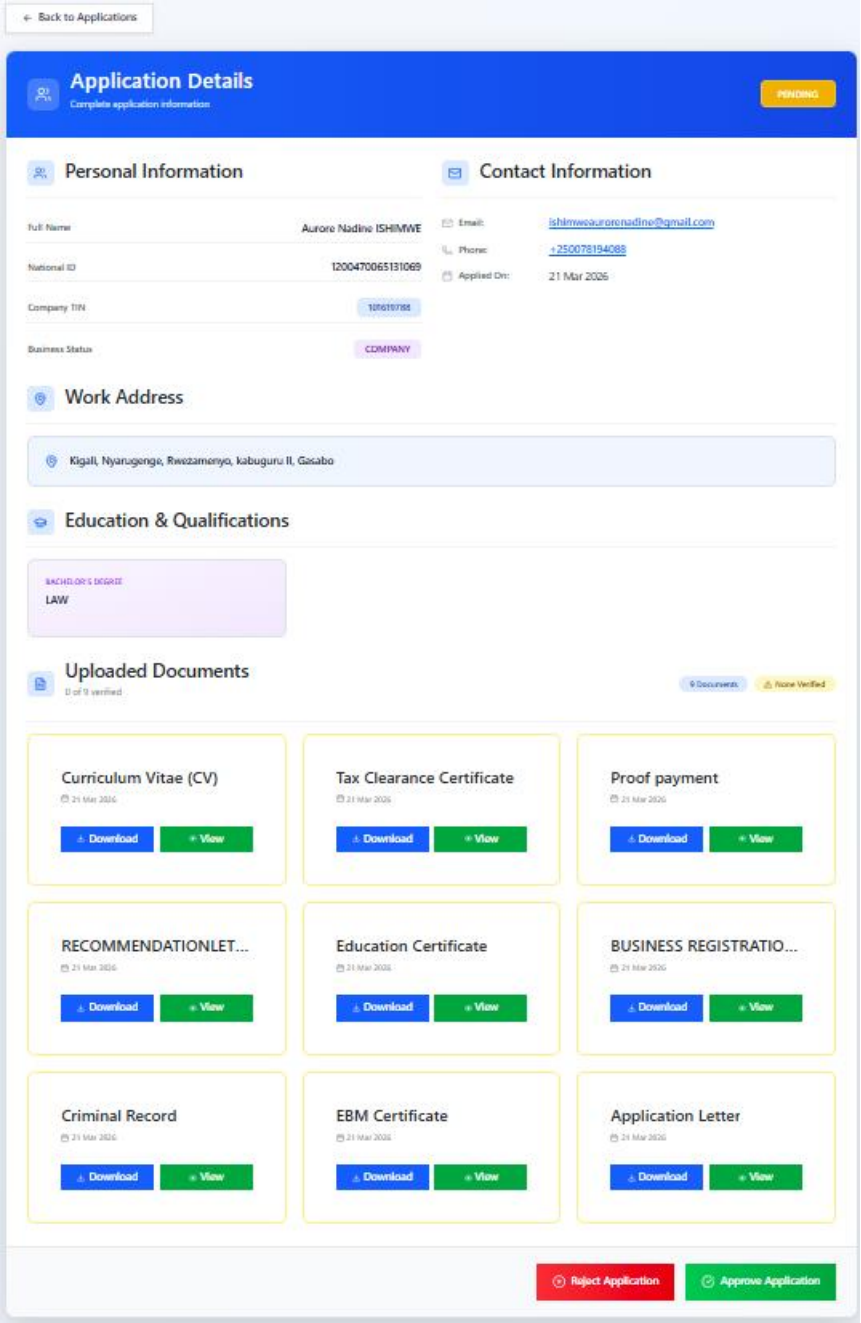


Figure 21: Application Detail Page

Software Testing

Unit Testing

Each module, including authentication, application submission, document upload, and approval processing, was tested separately to ensure correct functionality and proper validation of data.

Integration Testing

Testing was done to confirm smooth communication between the React frontend, Spring Boot backend, and PostgreSQL database, ensuring that data is correctly sent, stored, and retrieved.

Security Testing

Security testing verified that Spring Security with JWT correctly restricts access to authorized users only and protects the system from unauthorized access and invalid requests.

Usability Testing

Usability testing checked that applicants, officers, and administrators can easily use the system, confirming that the interface is clear, responsive, and easy to navigate.

Hardware and Software Requirements

Client-Side Requirements

The system can run on a standard computer with a modern web browser.

- **Operating System**: Windows 10 or later, macOS, or Linux
- **Processor**: Intel Core i3 or higher
- **RAM**: Minimum 4 GB (8 GB recommended)
- **Storage**: At least 1 GB free space
- **Browser**: Google Chrome, Firefox, or Edge (latest version)
- **Internet**: Stable internet connection for accessing the web system

Server-Side Requirements

The server hosts the backend application and database.

- **Operating System**: Windows Server or Linux (Ubuntu recommended)

- **Processor**: Intel Core i5 or higher
- **RAM**: Minimum 8 GB
- **Storage**: 20 GB or more
- **Backend**: Java 17 with Spring Boot
- **Database**: PostgreSQL

Deployment and System Configuration

The system is designed to run in a standard web server environment, with the React frontend running in a web browser and the Spring Boot backend connected to a PostgreSQL database. This setup ensures stable performance during development and testing while allowing future deployment on a cloud or larger server. The architecture supports easy maintenance, scalability, and reliable operation of the Tax Professional Management System.

Network and Security Requirements

The system uses secure configurations to protect user data and prevent unauthorized access. Authentication is handled using Spring Security with JWT, ensuring that only authorized users can access the system. All communication between the frontend and backend should use secure connections (HTTPS), and database access is protected using user roles and permissions. These measures help maintain data safety, system reliability, and proper access control in the Tax Professional Management System.

CHAPTER 5

CONCLUSIONS AND RECOMMENDATIONS

Conclusions

The development and implementation of the Tax Professional Management System represents an important improvement in the management of tax professional applications by replacing manual and paper-based processes with a digital solution. The system simplifies the submission, verification, and approval of applications, making the process faster, more accurate, and easier to manage. By providing a centralized platform for applicants, officers, and administrators, the system reduces errors, improves record keeping, and ensures better organization of application data.

The project shows how modern web technologies such as React, Spring Boot, PostgreSQL, and secure authentication using JWT can be used to build a reliable and secure management system. The system allows applicants to submit their applications online, enables officers to review and approve them efficiently, and ensures that all records are stored safely in the database. The modular design of the system makes it easy to maintain and allows future improvements without affecting existing features.

Testing confirmed that the system performs correctly, is easy to use, and provides secure access for different types of users. The authentication mechanism protects user data, while the structured database ensures that information is stored and retrieved correctly. Overall, the Tax Professional Management System successfully solves the problems of the previous manual system by improving efficiency, accuracy, and transparency in the application management process.

Recommendations

The Rwanda Revenue Authority and related institutions should support the use of the Tax Professional Management System to replace manual application processing with a digital platform. Using the system will improve the efficiency of application submission, verification, and approval, while ensuring that records are stored safely and can be accessed easily when needed. Clear guidelines for data management should be established to make sure applicant information is handled securely and according to national data protection standards.

Training should also be provided to officers and administrators so that they can use the system correctly and efficiently. Continuous system maintenance and updates will help ensure reliable performance and allow the system to adapt to future requirements.

Government institutions should encourage the use of digital systems in managing professional registrations and other administrative services. Supporting secure online platforms will improve transparency, reduce paperwork, and make public services faster and more reliable. Investment in server infrastructure, cybersecurity, and database management is recommended to ensure that systems like the Tax Professional Management System operate safely and efficiently. Policies that promote digital transformation in public institutions will also help improve service delivery and data management.

Future development of the system can include additional features to make the platform more powerful and flexible. Possible improvements include adding online payment integration, advanced reporting tools, and better search and filtering for applications. The system can also be extended to support mobile access and integration with other government systems. Further research may focus on improving security, increasing system performance, and adding more automation to reduce manual work for officers and administrators.

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APPENDICES

Curriculum Vitae

AURORE NADINE ISHIMWE

Software Engineering Student

aurorenadine25@gmail.com | +250 781 940 884 | Kigali, Rwanda

PROFESSIONAL SUMMARY

Dedicated Software Engineering student with practical experience in software development and IT operations. Proven track record in sales management and currently gaining valuable experience in IT systems at Rwanda Revenue Authority. Skilled in modern web technologies and passionate about creating innovative software solutions.

EDUCATION

Bachelor of Science in Software Engineering
Adventist University of Central Africa (AUCA)
2023 - 2025

Advanced Level - PCM Combination
Groupe Scolaire Saint Bernadette Save
2019 - 2022

Ordinary Level
Groupe Scolaire Mère du Verbe Kibeho
2016 - 2018

WORK EXPERIENCE

IT Intern
Rwanda Revenue Authority (RRA) | *Present*

- Gaining hands-on experience in IT systems administration, support, and software implementation within a government institution

Sales Manager
Queen Auto and Spare Parts LTD | *2 Years*

- Managed sales operations, customer relations, and inventory management
- Developed strong communication and leadership skills while driving business growth

Software Development Trainee
CSR | *4 Months*

- Participated in software development projects, learning industry best practices and collaborative development methodologies

TECHNICAL SKILLS

Frontend Development: React.js, HTML, CSS, JavaScript, C#

Backend Development: Spring Boot, c#

Professional Skills: Communication, Leadership, Management, ICT Skills

LANGUAGES

- **Kinyarwanda:** Excellent
- **English:** Excellent
- **French:** Good

INTERESTS

Reading • Coding • Singing • Music

REFERENCES

Mukabideli Clarisse
CEO, Queen Auto and Spare Parts LTD
Tel: +250 788 598 603

Ivan Bisengimana
System Developer
Email: kiyengobis@gmail.com

I, Aurore Nadine Ishimwe, declare that the information above is true to the best of my knowledge.

Data collection letter

Approval Letter

Rwanda Revenue Authority
IT Department
Application Unit
Kigali, Rwanda

January 23, 2026

TO WHOM IT MAY CONCERN,

RE: ACCEPTANCE OF ACADEMIC PROJECT FOR DATA COLLECTION/ACADEMIC PURPOSES

The Rwanda Revenue Authority (RRA) hereby acknowledges receipt and review of the academic project proposal submitted by Ms. ISHIMWE Aurore Nadine, a final-year student, entitled "**Tax Professionals Management System**," and hereby confirms that she is permitted to proceed with the implementation of the project in line with Rwanda Revenue Authority requirements.

The project aims to address challenges related to the management and regulation of tax professionals, including the application, review, approval, and monitoring processes for individuals seeking to operate as tax advisors. The system is designed as a web-based application that enables prospective tax advisors to submit applications and required documentation, while allowing authorized tax officers to review, approve, or reject applications in an efficient and transparent manner.

Application unit (RRA) hereby grants approval for the development and implementation of this tax professional is management system. We believe this project supports our digital transformation goals and will enhance tax advisors' experience.

Digitally signed by
trust_c1fd3c61-75cd-42
29-b4f7-547aa5c06b55
Date: 2026.01.25
06:01:08 +02'00'

Yours faithfully,